Article

Uncertainty-Aware Temporal Convolutional Networks for Multivariate Anomaly Detection: A Composite-Objective Framework with Chebyshev Bounds

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Abstract

Multivariate time-series anomaly detection on physical sensor networks faces three challenges that generic deep learning models inadequately addressed: heterogeneous sensor reliability, context-dependent anomaly scoring, and inactionable binary outputs lacking per-sensor attribution. We propose an uncertainty-aware Temporal Convolutional Network (TCN) framework built on two tightly integrated uncertainty-driven components: (i) an Adaptive Uncertainty-Aware Attention (AUAA) mechanism that gates temporal attention weights by per-sensor predictive uncertainty obtained from Monte-Carlo dropout; and (ii) a Dynamic Weight Adapter that learns context-sensitive blending of reconstruction error and uncertainty via a GRU over weight history. The architecture also includes an exploratory per-sensor attribution head, which we audit rather than claim: a controlled-perturbation test shows it is not yet causally faithful. We complement the empirical architecture with two distribution-free theoretical results: a Chebyshev-type false-positive bound on the hybrid anomaly score, and a Monte-Carlo posterior moment convergence result at rate $O(M^{-1/2})$. Evaluated on four-month indoor air quality sensor data, the Full Enhanced model achieves $R^2 = 0.9988$ and MSE 1.65×10^{-4}, a 25.2% MSE reduction over the Base TCN ($R^2 = 0.9984$, MSE 2.20×10^{-4}). Because the IAQ stream is unlabelled, the primary quantitative detection evaluation uses the labelled Skoltech Anomaly Benchmark (SKAB), a publicly available industrial water-circulation corpus disjoint from the IAQ training distribution; it yields an 8.8× F1 advantage (0.477 vs. 0.054) and a 14.4× recall advantage (0.418 vs. 0.029) for the proposed model configuration over the Base TCN at a validation-calibrated threshold applied without retuning. Against twelve established detectors under a unified protocol the proposed model attains the best F1 and recall, while the strongest reconstruction baselines retain higher precision and a marginally higher ROC-AUC, a recall-driven trade-off. Ablation isolates each component's contribution, the detector degrades gracefully under channel masking and noise, and the distribution-free false-positive bound is empirically respected. The framework retains a low inference cost (0.16 ms per window at $M = 20$ Monte-Carlo samples, including the uncertainty pass).

Keywords: multivariate time series; anomaly detection; temporal convolutional network; uncertainty quantification; Monte-Carlo dropout; adaptive attention; Chebyshev

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1. Introduction

Multivariate time-series data generated by sensor networks for environmental monitoring, industrial control, and building management embed critical operational intelligence and pose distinct analytical challenges that drive recent advances in time-series analysis [1,2]. Among these challenges, anomaly detection on multivariate time series is a central problem with applications spanning fault diagnosis, structural health monitoring, and process safety. Indoor air quality (IAQ) monitoring exemplifies the multivariate sensor-stream system: CO_2, volatile organic compounds (VOC), temperature, humidity, and particulate matter exhibit complex co-variation patterns while each sensor type suffers from distinct noise characteristics, calibration drift rates, and failure modes [3,4]. Time-series anomaly detection has been extensively surveyed [5–7], with recent IAQ-specific studies leveraging deep architectures for both forecasting and outlier identification [8,9]. In practice, deployment necessitates solutions that confront challenges unique to physical sensor networks, situated within a mathematically rigorous time-series framework challenges that conventional time-series models fail to explicitly address.

This study is motivated by three sensor-specific challenges. First, multi-sensor deployments exhibit heterogeneous reliability across time. Individual sensors degrade at different rates: electrochemical VOC sensors drift over weeks, optical dust sensors saturate in high-concentration events, and temperature sensors maintain stable calibration for months [10]. Standard attention mechanisms treat all time steps equally, yet predictions for a degrading VOC sensor at time t are inherently less reliable than predictions for a stable temperature sensor at the same instant. A sensor-aware model should dynamically down-weight attention to unreliable channels. Second, anomaly scoring must adapt to context. Most approaches combine reconstruction error and prediction uncertainty using fixed weights [11,12], but the relative informativeness of these signals depends on sensor state: during stable operation, large reconstruction errors reliably indicate anomalies; During degradation events, a high level of uncertainty may actually serve as a more reliable signal, even though reconstruction errors appear moderate. A fixed-weight scoring system combines these two distinct conditions, thereby reducing detection accuracy during sensor transitions. Third, anomaly detectors should produce actionable output. Binary flagging without sensor-level attribution leaves facility operators unable to determine whether a detected anomaly reflects a genuine environmental hazard, a degrading CO_2 sensor requiring recalibration, or a VOC spike from cleaning activities [13]. For anomaly detection to inform maintenance decisions, it must identify which sensors contribute to each anomaly and provide uncertainty-aware estimates rather than only point predictions.

The Temporal Convolutional Network (TCN) architecture [14] provides an effective backbone for sequence modeling through dilated causal convolutions, matching or exceeding recurrent models on time-series benchmarks. Monte Carlo (MC) dropout [15] enables uncertainty estimation from deterministic architectures without requiring explicit Bayesian inference. Neither technique alone, however, addresses the sensor-specific challenges described above. The principal contribution of this study lies in the integration of uncertainty signals, not as independent outputs, but as internal mechanisms that actively influence attention computation, anomaly scoring, and sensor-level attribution, thereby aligning with the operational realities of physical sensor deployments.

We present the TCN framework, which accounts for uncertainty in the detection of multivariate sensor anomalies, and which makes three contributions closely related to the challenges outlined above:

1. **Adaptive Uncertainty-Aware Attention (AUAA).** An attention mechanism that modulates temporal attention weights using per-sensor predictive uncertainty estimates via an uncertainty gate $g_t = 1/(1+\sigma_t)$. This makes attention computation sensitive to sensor-specific reliability fluctuations, enabling the model to down-weight unreliable channels.
2. **Dynamic Weight Adapter for Hybrid Anomaly Scoring.** A learnable gating module that contextually balances reconstruction error against predictive uncertainty using a GRU over recent weight history. This replaces static combination weights with a mechanism that adapts to sensor operating conditions.
3. **Distribution-Free Operating Guarantees.** A Chebyshev-type false-positive bound that guarantees $P(\text{false positive}) \leq 1/k^2$ without distributional assumptions, together with a Monte-Carlo posterior-moment convergence result that fixes the sample budget M.

The architecture also includes a per-sensor attribution head; we treat it as *exploratory* and audit in Section 4.7 , and the detection and uncertainty results do not depend on it.

2. Related Works

2.1. Deep Anomaly Detection in Multivariate Time Series

Multivariate time-series anomaly detection has evolved from classical statistical methods toward deep architectures that learn normal behavioural patterns and identify deviations [7]. Foundational sequence models LSTM [16], GRU [17], and the TCN [14] established the architectural vocabulary for temporal modelling. The TCN, in particular, demonstrated that dilated causal convolutions provide a competitive alternative to recurrence, with advantages in training stability and parallel computation. WaveNet [18] originally introduced dilated causal convolutions for sequence modelling; subsequent comprehensive reviews [19] catalogue the diverse extensions that have since been applied to industrial time series.

Recent work has extended TCNs with complementary techniques: TCN-AE architectures [20] leverage autoencoding for unsupervised detection; TCN-Linear hybrids [21] combine convolutional and linear residual paths; and attention-augmented TCNs [22] incorporate self-attention for industrial monitoring applications. Reconstruction-based anomaly detectors such as USAD [23] and LSTM-ED [24] learn normal-behaviour priors and threshold residuals at inference. Transformer-based detectors such as TranAD [25] and Anomaly Transformer [13] introduce self-attention for anomaly scoring; recent designs (e.g., Informer [26], TimesNet [27], MTAD-GAT [28]) further enrich the attention vocabulary but do not incorporate predictive uncertainty into the attention mechanism itself.

Despite this progress, existing TCN-based detectors treat sensor readings as generic multivariate inputs without modelling sensor-specific reliability characteristics. Our work explicitly couples uncertainty estimation to attention computation and anomaly scoring, adapting model behaviour to the varying reliability of individual sensor channels. Uncertainty quantification (UQ) transforms anomaly detection from threshold-based heuristics into confidence-aware decisions [29]. Dropout as a Bayesian approximation [15] provides a lightweight mechanism for estimating predictive uncertainty from deterministic architectures, building on the original dropout regularizer [30]; deep ensembles [31] offer an alternative with stronger calibration at higher computational cost. MC dropout has been validated across diverse time-series domains: electricity demand forecasting with TCNs [11], stochastic differential equation-based hydrological forecasting [12], and multivariate imputation. These works demonstrate that MC dropout provides meaningful uncertainty estimates, but they apply uncertainty post hoc as a separate scoring term rather than feeding it back into the model's internal representations. Our framework advances this field of study by closing the loop: uncertainty estimates modulate attention weights and condition the dynamic scoring function, creating integration between UQ and model computation motivated by sensor-specific reliability concerns.

For attention mechanisms in temporal data, the Transformer's self-attention [32] enables the direct calculation of relationships between tokens and captures long-range dependencies that might be missed by convolutional filters. Standard attention assumes all input positions are equally trustworthy an assumption violated by physical sensor data where individual measurements carry sensor-specific uncertainty. Conformal prediction [33] and concentration-inequality machinery [34] provide distribution-free statistical guarantees, but operate at the output level rather than altering internal representations. Our AUAA mechanism integrates uncertainty directly into attention computation, enabling the model to learn which temporal patterns deserve attention based on the estimated reliability of the underlying sensor measurements; the resulting hybrid score is in turn bounded by a Chebyshev-type inequality (Theorem 1), providing the framework an explicit distribution-free false-positive guarantee. Multi-task uncertainty weighting [35] offers a related precedent for learning task-specific scalings, which our Dynamic Weight Adapter extends into a context-dependent (per-window) scheme. Interpretability remains a critical gap in practical anomaly detection. Methods such as SHAP and LIME provide post-hoc explanations but add computational overhead. Our feature-importance learner is trained jointly with the detection model, producing sensor-specific attributions as a native output without additional post-processing.

2.2. Benchmarks and Cross-Dataset Evaluation for Time-Series Anomaly Detection

The benchmarking practice in multivariate-TSAD has been a source of active concern. Wu and Keogh [36] document that several widely cited benchmarks exhibit triviality, mislabelling, and run-length bias that inflate reported scores; Lai et al. [37] consolidate a formal taxonomy of point, contextual, and collective anomalies and propose evaluation protocols that disentangle these regimes. Together, these works argue for evaluating detectors on (i) labelled corpora authored by domain experts rather than by the model developers themselves and (ii) at least one dataset outside the training distribution. Hundman et al. [38] provide the labelled NASA SMAP/MSL spacecraft telemetry corpus; Su et al. [39] provide the SMD server-machine corpus; Goh et al. [40] and Mathur and Tippenhauer [41] describe the SWaT water-treatment testbed; and Filonov, Lavrentyev, and Vorontsov [42] authored the predecessor of the SKAB testbed [43] used in our cross-dataset experiment (Section 4.5). We follow the cross-dataset protocol recommended by these critiques: the labelled evaluation is run on a corpus (SKAB) whose ground truth is exogenous to our model development pipeline and whose domain (industrial water circulation) is disjoint from the IAQ training distribution.

Table 1. Comparison of the proposed framework with existing approaches across key dimensions.

Method	Uncertainty Quant.	Adaptive Attention	Sensor Adaptation	Feature Attribution
TCN-AE [20]	No	No	No	No
LSTM-ED [24]	No	No	No	No
USAD [23]	No	No	No	No
Anomaly Transformer [13]	No	No	No	No
Stochastic TCN [11]	Yes	No	No	No
SDE-HNN [12]	Yes	No	No	No
This work	**Yes**	**Yes**	**Yes**	Partial†

† An attribution head is implemented but a controlled-perturbation audit (Section 4.7) shows it is not yet causally faithful; it is reported as an exploratory component, not a validated capability.

3. Methodology

3.1. Problem Statement and Sensor-Specific Formulation

Consider a multivariate sensor network with n sensors deployed in a physical environment, producing observations $\mathbf{x}_t \in \mathbb{R}^n$ at each discrete time step t. Each sensor $f \in \{1, \ldots, n\}$ generates a scalar reading $x_f(t)$ with sensor-specific noise characteristics:

$$x_f(t) = \mu_f(t) + \eta_f(t), \quad \eta_f(t) \sim \mathcal{N}(0, \sigma_f^2(t)) \tag{1}$$

where $\mu_f(t)$ is the true physical quantity and $\sigma_f^2(t)$ is the time-varying sensor-specific noise variance that may fluctuate due to aging, environmental interference, or calibration drift. Three sensor-specific challenges arise from this formulation:

1. **Heterogeneous noise profiles.** Different sensor technologies exhibit distinct noise characteristics: CO_2 sensors suffer low-frequency drift from optical path contamination, VOC sensors display sensitivity degradation over weeks, and optical dust sensors produce transient saturation spikes during high-concentration events. A model that treats all sensor channels identically cannot adapt to these per-sensor reliability differences.
2. **Dynamic cross-dependencies.** Multivariate sensor readings co-vary through shared environmental drivers (HVAC cycling, occupancy patterns, diurnal cycles), creating complex inter-sensor correlations that evolve over time.
3. **Actionable output requirements.** Facility operators need per-sensor anomaly attribution to distinguish genuine environmental hazards (e.g., a VOC leak) from sensor malfunctions (e.g., a drifting CO_2 sensor), enabling targeted maintenance responses.

The goal is to learn a function $f_\theta : \mathbb{R}^{W \times n} \to \mathbb{R}^n$ that maps a window of W past observations to next-step predictions, while jointly producing per-sensor uncertainty estimates, context-adaptive anomaly scores, and feature-level attributions.

3.2. Feature Engineering and Preprocessing

We construct a windowed feature representation for each sensor f at time t using a sliding window of length W:

$$\mathbf{h}_f(t) = \left[x_f(t), x_f(t-1), \ldots, x_f(t-W+1), \Delta_f(t), \mu_f^W(t), \sigma_f^W(t) \right] \tag{2}$$

where $\Delta_f(t) = x_f(t) - x_f(t-1)$ captures first-order dynamics, $\mu_f^W(t)$ is the rolling mean, and $\sigma_f^W(t)$ is the rolling standard deviation over the window. Overlapping windows prevent boundary detection gaps.

DBSCAN-based Outlier Filtering. Raw sensor data contains saturation artifacts (e.g., value 9999 for CO_2 and VOC sensors during calibration cycles or communication errors). We apply DBSCAN clustering [44] with neighborhood radius $\varepsilon = 0.5$ and minimum cluster size MinPts $= 10$, selected via grid search over $\varepsilon \in [0.1, 2.0]$ and MinPts $\in [5, 25]$ on the training set, to identify and remove these noise points while preserving the core data distribution. Table 2 summarizes the statistical characteristics before and after filtering.

Table 2. CO_2 and VOC statistics before and after DBSCAN-based saturation filtering, computed directly from the dataset (`filtered_data.csv` before; `df_normal.csv` after).

Statistic	Before Filtering		After Filtering	
	CO_2 (ppm)	VOC (ppb)	CO_2 (ppm)	VOC (ppb)
Count	178,560	178,560	178,560	178,560
Mean	246.5	573.9	236.4	558.6
Std	258.2	313.9	153.3	143.5
Min	0	400	0	400
Median	213	538	213	538
Max	9999	9999	2606	2725

3.3. Uncertainty-Calibrated TCN Architecture

The overall architecture of the proposed framework is illustrated in Figure 1. The framework follows a closed-loop design: sensor data pass through a three-layer TCN backbone, MC dropout produces per-sensor uncertainty estimates, and these uncertainty signals are then fed back into the attention mechanism (AUAA), the dynamic anomaly scoring (Weight Adapter), and the per-sensor explanation module (Feature Attribution) rather than being used only as an output-level term.

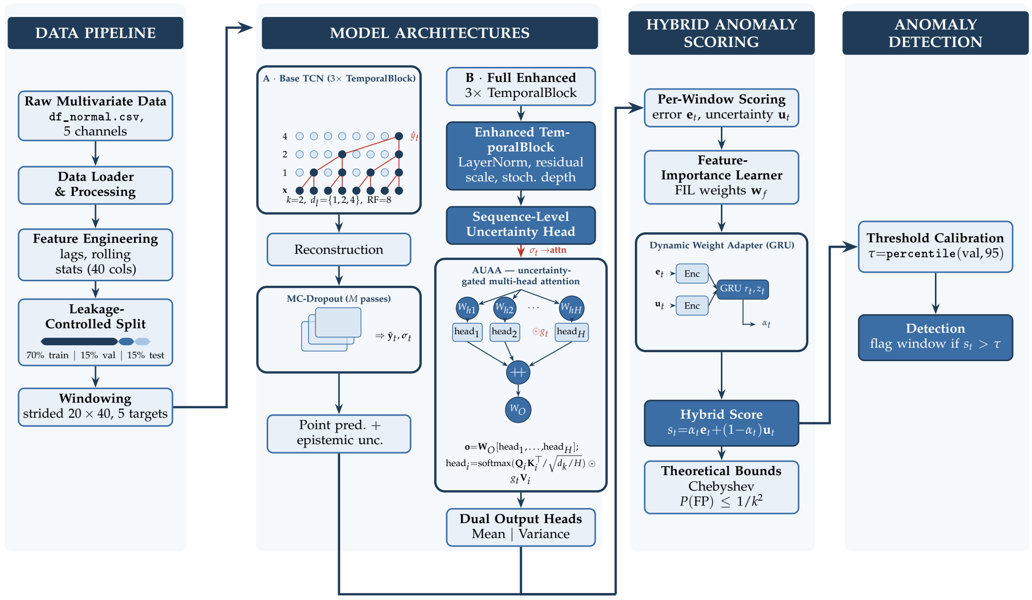


Figure 1. End-to-end architecture of the proposed uncertainty-aware TCN framework as a four-stage pipeline: Data Pipeline → Model Architectures (Base TCN and Full Enhanced) → Hybrid Anomaly Scoring → Anomaly Detection. Inset breakdowns show the dilated causal convolution stack (k=2, d_l=$\{1,2,4\}$, RF=8), MC-dropout inference, the uncertainty-gated AUAA (g_t=$1/(1+\sigma_t)$), and the GRU Dynamic Weight Adapter (α_t). Dark boxes mark contributed components; the dashed red arrow marks the uncertainty-to-attention coupling.

Intuition. Before the formal treatment, here is how the uncertainty signal travels through the model. The TCN first produces a prediction; repeating this under MC dropout turns the spread of the forward passes into a per-channel predictive uncertainty σ_t. This single quantity is then reused in three ways: to *down-weight* attention at time steps the model is unsure about (AUAA), to *decide* whether the final anomaly score should lean more on the reconstruction error or on the uncertainty (Dynamic Weight Adapter), and to *flag* an anomaly only when the combined score crosses a threshold that carries a provable false-alarm ceiling (Chebyshev bound). The rest of this section formalizes these three uses; a reader interested in the operational behaviour rather than the derivations may skip ahead to the hybrid score in Equation (17).

TCN Backbone. The core of our framework is a three-layer Temporal Convolutional Network [14,18] with 64 channels per layer, kernel size $k = 2$, and exponentially increasing dilation factors $d_l = 2^{l-1}$ for $l \in \{1, 2, 3\}$, yielding $\{d_1, d_2, d_3\} = \{1, 2, 4\}$. Within layer l, each output time step is produced by a dilated causal convolution:

$$y_t^{(l)} = \sum_{i=0}^{k-1} f^{(l)}(i) \cdot x_{t-d_l \cdot i}^{(l-1)}, \tag{3}$$

where $f^{(l)}(\cdot)$ are the layer-l kernel weights and $x_{t-d_l \cdot i}^{(l-1)}$ is the input at the corresponding past time step. The closed-form receptive field of the stack is

$$RF = 1 + (k-1)\sum_{l=1}^{L} d_l = 1 + (k-1)(2^L - 1) = 8 \tag{4}$$

with $L = 3$ layers, covering 8 of the $W = 20$ input timesteps; this matches the temporal scale at which IAQ co-variation manifests (sub-10-minute occupancy and ventilation transients) without inflating the parameter count. Each layer applies causal (left-padded) convolution to prevent information leakage from future timesteps, followed by weight normalization and ReLU activation.

Stochastic Depth Integration. During training, each TCN block l is randomly skipped via a stochastic-depth Bernoulli gate [45]:

$$\mathbf{x}^{(l+1)} = \mathbf{x}^{(l)} + b_l \cdot F^{(l)}(\mathbf{x}^{(l)}), \quad b_l \sim \text{Bernoulli}(1 - p_l), \tag{5}$$

where $F^{(l)}$ is the residual function of block l and $p_l = 0.1 \times (l/L)$ scales linearly with depth. At inference, all blocks participate deterministically (i.e. $b_l = 1$), and Monte Carlo dropout is activated across all layers with dropout rate $r = 0.2$.

MC Dropout as Bayesian Posterior. Following [15], dropout applied at inference approximates the posterior predictive distribution:

$$p(\hat{\mathbf{y}}_t \mid \mathbf{x}_t, \mathcal{D}) \approx \frac{1}{M} \sum_{m=1}^{M} p\Big(\hat{\mathbf{y}}_t \mid \mathbf{x}_t, \boldsymbol{\theta}^{(m)}\Big), \tag{6}$$

where $\boldsymbol{\theta}^{(m)}$ are the parameters with random dropout masks at the m-th forward pass. Running M stochastic forward passes yields an ensemble $\{\hat{\mathbf{y}}^{(m)}\}_{m=1}^{M}$ from which we compute the predictive mean and the per-sensor predictive standard deviation:

$$\hat{\mathbf{y}}_t = \frac{1}{M} \sum_{m=1}^{M} \hat{\mathbf{y}}_t^{(m)}, \quad \sigma_t = \sqrt{\frac{1}{M} \sum_{m=1}^{M} (\hat{\mathbf{y}}_t^{(m)} - \hat{\mathbf{y}}_t)^2}. \tag{7}$$

The variance σ_t^2 admits an epistemic–aleatoric decomposition in the heteroscedastic case:

$$\sigma_t^{2,\text{total}} = \underbrace{\frac{1}{M}\sum_{m=1}^{M}(\hat{\mathbf{y}}_t^{(m)} - \hat{\mathbf{y}}_t)^2}_{\text{epistemic}} + \underbrace{\frac{1}{M}\sum_{m=1}^{M}\hat{\sigma}_t^{(m),2}}_{\text{aleatoric}}, \tag{8}$$

where $\hat{\sigma}_t^{(m)}$ are per-pass aleatoric variance estimates when the network has a heteroscedastic output head. In our experiments the homoscedastic case ($\hat{\sigma}_t^{(m)} = 0$) is used, so $\boldsymbol{\sigma}_t$ in Equation (7) captures only the epistemic component.

3.4. Adaptive Uncertainty-Aware Attention (AUAA)

The AUAA mechanism modulates standard scaled dot-product attention with per-timestep predictive uncertainty. Given query $\mathbf{Q}_t$, key $\mathbf{K}_t$, and value $\mathbf{V}_t$ matrices of dimension d_k, the uncertainty-gated attention is:

$$\boldsymbol{\alpha}_t = \text{softmax}\left(\frac{\mathbf{Q}_t\mathbf{K}_t^T}{\sqrt{d_k}}\right) \odot \frac{1}{1+\boldsymbol{\sigma}_t} \tag{9}$$

where $\odot$ denotes element-wise multiplication and $\boldsymbol{\sigma}_t$ is the predictive standard deviation vector from MC dropout. The uncertainty gate $g_t = 1/(1+\sigma_t)$ maps high-uncertainty predictions ($\sigma_t \to \infty$) to near-zero attention weights and low-uncertainty predictions ($\sigma_t \to 0$) to full attention. A learnable sigmoid refinement layer provides additional flexibility:

$$g_t^{\text{learned}} = \text{sigmoid}(\mathbf{W}_g \cdot \boldsymbol{\sigma}_t + \mathbf{b}_g) \tag{10}$$

where $\mathbf{W}_g$ and $\mathbf{b}_g$ are the learnable weight matrix and bias vector of the refinement layer and sigmoid$(\cdot)$ is applied element-wise.

The AUAA module uses $H = 4$ attention heads with hidden dimension 64, with head outputs concatenated and linearly projected. The multi-head extension applies the uncertainty gate independently per head:

$$\boldsymbol{\alpha}_t^{(h)} = \text{softmax}\left(\frac{\mathbf{Q}_t^{(h)}\mathbf{K}_t^{(h)\top}}{\sqrt{d_k/H}}\right) \odot \frac{1}{1+\boldsymbol{\sigma}_t}, \quad h = 1,\ldots,H, \tag{11}$$

followed by a linear projection $\mathbf{W}_O\,[\boldsymbol{\alpha}_t^{(1)},\ldots,\boldsymbol{\alpha}_t^{(H)}]$, where $\mathbf{W}_O$ is the learnable output-projection matrix, H the number of heads, d_k/H the per-head dimension, and the superscript (h) indexes the h-th head. This uncertainty-gated attention differs fundamentally from standard attention plus dropout regularisation: dropout perturbs attention randomly with no awareness of prediction quality, while AUAA explicitly couples attention weights to the model's own estimate of per-sensor reliability. Figure 2 illustrates the dataflow that distinguishes AUAA from a standard scaled dot-product attention block.

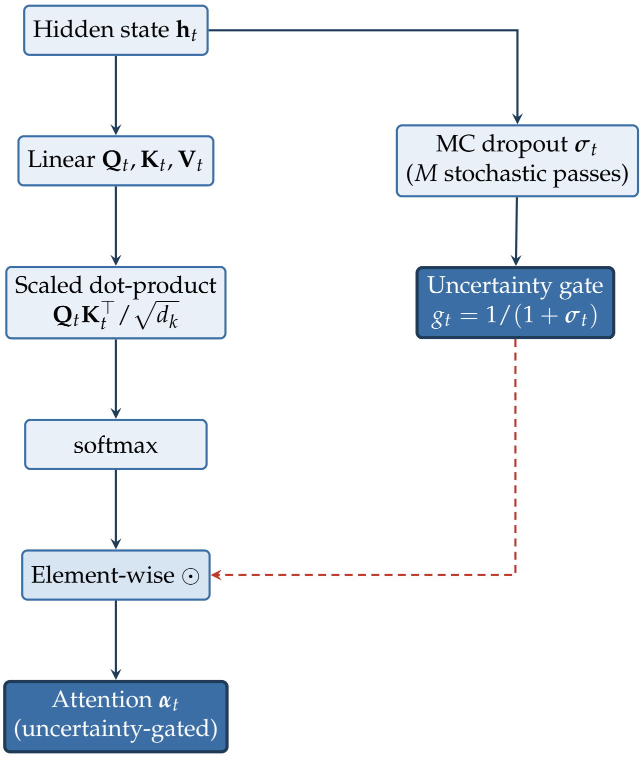


Figure 2. Adaptive Uncertainty-Aware Attention (AUAA) dataflow. Standard scaled dot-product attention (left) is element-wise modulated by an uncertainty gate $g_t = 1/(1+\sigma_t)$ from MC dropout (right); the dashed red line marks the uncertainty feedback path.

3.5. Dynamic Weight Adapter for Anomaly Scoring

Fixed-weight combination of reconstruction error and uncertainty fails to capture context-dependent sensor behavior. Our Dynamic Weight Adapter learns time-varying weights conditioned on both signals and their history. At each timestep, reconstruction error $\mathbf{e}_t = |\mathbf{x}_t - \hat{\mathbf{x}}_t|$ and predictive uncertainty $\mathbf{u}_t = \sigma_t$ are concatenated with a context vector $\mathbf{h}_t$:

$$\alpha_t = \text{sigmoid}(\mathbf{W}_\alpha \cdot [\mathbf{e}_t, \mathbf{u}_t, \mathbf{h}_t] + \mathbf{b}_\alpha) \tag{12}$$

where $\mathbf{W}_\alpha$ and $\mathbf{b}_\alpha$ are learned parameters, $[\cdot,\cdot,\cdot]$ denotes concatenation, $\mathbf{e}_t$ is the reconstruction-error vector, $\mathbf{u}_t = \sigma_t$ the predictive-uncertainty vector, and $\mathbf{h}_t$ is the hidden state of a GRU that tracks recent weight history, enabling the adapter to remember past sensor reliability trends. The GRU recurrence is given explicitly by

$$\mathbf{r}_t = \text{sigmoid}(\mathbf{W}_r[\mathbf{e}_t, \mathbf{u}_t] + \mathbf{U}_r\mathbf{h}_{t-1} + \mathbf{b}_r), \tag{13}$$
$$\mathbf{z}_t = \text{sigmoid}(\mathbf{W}_z[\mathbf{e}_t, \mathbf{u}_t] + \mathbf{U}_z\mathbf{h}_{t-1} + \mathbf{b}_z), \tag{14}$$
$$\tilde{\mathbf{h}}_t = \tanh(\mathbf{W}_h[\mathbf{e}_t, \mathbf{u}_t] + \mathbf{U}_h(\mathbf{r}_t \odot \mathbf{h}_{t-1}) + \mathbf{b}_h), \tag{15}$$
$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t, \tag{16}$$

where $\mathbf{r}_t$ and $\mathbf{z}_t$ are the reset and update gates, $\tilde{\mathbf{h}}_t$ the candidate state, $\{\mathbf{W}_r, \mathbf{W}_z, \mathbf{W}_h\}$ and $\{\mathbf{U}_r, \mathbf{U}_z, \mathbf{U}_h\}$ learned weight matrices, $\{\mathbf{b}_r, \mathbf{b}_z, \mathbf{b}_h\}$ bias vectors, and $\odot$ the Hadamard (element-wise) product. The hybrid anomaly score combines both signals:

$$s_t = \alpha_t \cdot \mathbf{e}_t + (1 - \alpha_t) \cdot \mathbf{u}_t \tag{17}$$

During stable sensor operation, $\alpha_t \to 1$, making the score reconstruction-error dominant; during sensor degradation, $\alpha_t \to 0$, shifting emphasis to uncertainty. This context-sensitive mechanism adapts to evolving sensor conditions without manual threshold tuning.

3.6. Feature-Level Anomaly Attribution

The feature-importance learner computes per-sensor attribution weights via a learned transformation:

$$\mathbf{w}_f = \mathrm{softmax}(\mathbf{V} \cdot \tanh(\mathbf{W} \cdot \mathbf{x} + \mathbf{b})) \tag{18}$$

where $\mathbf{W} \in \mathbb{R}^{d\times n}$ and $\mathbf{V} \in \mathbb{R}^{1\times d}$ are learned weight matrices, $\mathbf{b} \in \mathbb{R}^d$ a learned bias vector, d the hidden width, n the number of sensors, $\mathbf{x}$ the input window, and $\mathbf{w}_f$ the resulting per-sensor attribution weights. An inter-sensor correlation matrix $\mathbf{C} \in \mathbb{R}^{n\times n}$ is also learned to capture physical cross-sensor couplings (e.g., HVAC-induced correlations between temperature, humidity, and CO_2).

Theoretical Guarantee (Chebyshev Bound). Let s_t be the anomaly score at time t with empirical mean $\bar{s}$ and standard deviation σ_s over a calibration set. For any threshold $\tau = \bar{s} + k \cdot \sigma_s$ with $k > 0$, the Chebyshev concentration inequality provides a distribution-free bound on the false positive rate:

$$P(\text{false positive}) = P(s_t > \tau \mid \text{normal}) \leq \frac{1}{k^2} \tag{19}$$

This guarantees that for a desired false positive rate $\epsilon = 0.05$, setting $k = \sqrt{1/\epsilon} = \sqrt{20} \approx 4.47$ ensures the actual false positive rate does not exceed 5%, regardless of the underlying score distribution. This is a strictly stronger guarantee than percentile-based thresholds, which provide only empirical estimates with no statistical coverage.

3.7. Composite Loss Function

The model is trained with a composite objective that jointly optimizes reconstruction fidelity, uncertainty calibration, and weight stability:

$$\mathcal{L} = \mathcal{L}_{\text{MSE}} + \beta_1 \cdot \mathcal{L}_{\text{uncertainty}} + \beta_2 \cdot \mathcal{L}_{\text{weight}} \tag{20}$$

where $\mathcal{L}_{\text{MSE}} = \frac{1}{n}\sum_{f=1}^{n}(\hat{y}_f - y_f)^2$ is the reconstruction loss. The uncertainty calibration term $\mathcal{L}_{\text{uncertainty}} = \frac{1}{n}\sum_{f=1}^{n}(\sigma_f^{\text{pred}} - \sigma_f^{\text{emp}})^2$ penalizes discrepancies between the predicted standard deviation σ_f^{pred} (from MC dropout) and the empirical residual standard deviation σ_f^{emp} computed over a rolling window of 50 samples, discouraging overconfident predictions. The weight regularization term $\mathcal{L}_{\text{weight}} = \frac{1}{T}\sum_{t=2}^{T}(\alpha_t - \alpha_{t-1})^2$ encourages smooth temporal transitions in the dynamic weight α_t, preventing erratic shifts between reconstruction- and uncertainty-dominated regimes. Coefficients β_1 and β_2 converged to $\beta_1 = 0.1$ and $\beta_2 = 0.05$ via grid search on the validation set, with β_2 set lower to avoid over-smoothing the dynamic weights

3.8. Anomaly Detection Pipeline

We implement four thresholding strategies:

1. **Percentile-based** (*empirical*):

$$\tau_{95} = Q_{0.95}\Big(\{s_t^{\text{val}}\}_{t=1}^{T_{\text{val}}}\Big), \tag{21}$$

where $Q_{0.95}(\cdot)$ is the 95^{th} percentile of validation-set anomaly scores.

2. **Adaptive** (*context-aware*):

$$\tau_{\text{adapt}} = \tau_{\text{base}} \cdot (0.8 + 0.4 \cdot \bar{\alpha}_t), \tag{22}$$

where $\bar{\alpha}_t$ is a moving average of the dynamic weight from Equation (12) over a local window.
3. **Chebyshev theoretical** (*distribution-free FP guarantee*):

$$\tau_{\text{theory}} = \bar{s} + k \cdot \sigma_s, \quad k = \sqrt{1/\epsilon}, \quad P(\text{FP}) \leq \frac{1}{k^2} = \epsilon, \tag{23}$$

where $\bar{s}, \sigma_s$ are the mean and standard deviation of validation scores and ϵ is the desired false-positive rate. The formal guarantee is stated in Theorem 1 below.
4. **Distribution-free quantile** (*contamination-aware*):

$$\tau_{\text{dist}} = Q_{1-\gamma}(\hat{F}_s), \tag{24}$$

where $\hat{F}_s$ is the empirical CDF of the hybrid score s_t from Equation (17) and γ is the assumed contamination ratio.

The complete detection algorithm is presented in Algorithm 1.

3.9. Theoretical Guarantees

The two results below formalize the distribution-free properties on which our anomaly-detection pipeline rests. Both are stated for the anomaly-score variable produced by Equation (17).

Theorem 1 (Distribution-Free False-Positive Bound)**.** *Let S denote the anomaly-score random variable on a clean (normal-operation) validation set, with finite mean $\mu_s = \mathbb{E}[S]$ and finite variance $\sigma_s^2 = \operatorname{Var}(S) < \infty$. For any desired false-positive rate $\epsilon \in (0, 1)$, define the threshold*

$$\tau_{\text{theory}}(\epsilon) \triangleq \mu_s + k(\epsilon)\, \sigma_s, \qquad k(\epsilon) = \sqrt{1/\epsilon}.$$

Then the false-positive rate of the detector $\mathbf{1}_{[S > \tau_{\text{theory}}(\epsilon)]}$ satisfies

$$P\Big(S > \tau_{\text{theory}}(\epsilon)\Big) \;\leq\; \frac{1}{k(\epsilon)^2} \;=\; \epsilon,$$

without any distributional assumption on S.

Proof. By Chebyshev's inequality [6], for any random variable S with finite mean μ_s and variance σ_s^2 and for every $k > 0$,

$$P(|S - \mu_s| \geq k\, \sigma_s) \;\leq\; \frac{1}{k^2}.$$

The one-sided event $\{S > \mu_s + k\sigma_s\}$ is contained in $\{|S - \mu_s| \geq k\sigma_s\}$, so $P(S > \mu_s + k\sigma_s) \leq P(|S - \mu_s| \geq k\sigma_s) \leq 1/k^2$. Setting $k = \sqrt{1/\epsilon}$ yields $1/k^2 = \epsilon$, which establishes the claimed bound.

Theorem 2 (Monte-Carlo Posterior Moment Convergence)**.** *Let $\boldsymbol{\theta}^{(m)} \overset{\text{i.i.d.}}{\sim} q(\boldsymbol{\theta})$ for $m = 1, \ldots, M$ denote dropout-sampled network weights, and define the per-sample predictions $\hat{\mathbf{y}}^{(m)}(\mathbf{x}) =$*

$f_{\boldsymbol{\theta}^{(m)}}(\mathbf{x})$. *Assume* $\hat{\mathbf{y}}^{(m)}$ *has finite second moment* $\mathbb{E}_q[\|\hat{\mathbf{y}}^{(m)}\|^2] < \infty$. *Then the empirical Monte-Carlo mean and variance estimators*

$$\hat{\mathbf{y}}_M \triangleq \frac{1}{M}\sum_{m=1}^{M}\hat{\mathbf{y}}^{(m)}, \qquad \hat{\sigma}^2_M \triangleq \frac{1}{M}\sum_{m=1}^{M}(\hat{\mathbf{y}}^{(m)} - \hat{\mathbf{y}}_M)^2$$

converge to the posterior predictive moments at rate $O(1/\sqrt{M})$ *in probability:* $\|\hat{\mathbf{y}}_M - \mathbb{E}_q[\hat{\mathbf{y}}]\| = O_p(M^{-1/2})$ *and likewise for* $\hat{\sigma}^2_M$.

Proof. We first fix the convergence notation. For a sequence of random vectors $\{Z_M\}$, $Z_M \xrightarrow{a.s.} Z$ denotes *almost-sure convergence* ($P(\lim_{M\to\infty} Z_M = Z) = 1$); $Z_M \xrightarrow{d} Z$ denotes *convergence in distribution* (pointwise convergence of the cumulative distribution functions at every continuity point of the limit); and $Z_M = O_p(a_M)$ denotes *stochastic boundedness at rate* a_M, i.e. for every $\varepsilon > 0$ there exist $C_\varepsilon, M_\varepsilon$ with $P(\|Z_M\|/a_M > C_\varepsilon) < \varepsilon$ for all $M \geq M_\varepsilon$. The strong law of large numbers implies $\hat{\mathbf{y}}_M \xrightarrow{a.s.} \mathbb{E}_q[\hat{\mathbf{y}}]$ since the i.i.d. samples have finite first moment. The classical central limit theorem then yields $\sqrt{M}\,(\hat{\mathbf{y}}_M - \mathbb{E}_q[\hat{\mathbf{y}}]) \xrightarrow{d} \mathcal{N}(0, \mathrm{Cov}_q(\hat{\mathbf{y}}))$ under finite second moment, giving the $O_p(M^{-1/2})$ rate. The same argument applied to the centred squared sample $(\hat{\mathbf{y}}^{(m)} - \hat{\mathbf{y}}_M)^2$ together with Slutsky's theorem gives convergence of $\hat{\sigma}^2_M$ at the same rate. The variational interpretation of dropout as a Bayesian posterior $q(\boldsymbol{\theta})$ over weights is due to Gal and Ghahramani [15]; under their identification, $\mathbb{E}_q[\hat{\mathbf{y}}]$ and $\mathrm{Cov}_q(\hat{\mathbf{y}})$ are the posterior predictive moments.

Theorem 1 furnishes the false-positive guarantee invoked at line 16 of Algorithm 1. Theorem 2 justifies the empirical moments used at lines 4–5 of the same algorithm, with the rate $M = 50$ chosen so that $M^{-1/2} \approx 0.14$ for an a-priori standard-error budget under 15%. With the architectural components, detection pipeline, and theoretical guarantees fully specified, the following section evaluates each component's contribution through forecasting benchmarks, ablation studies, sensitivity analyses, and a cross-dataset validation on the Skoltech Anomaly Benchmark.

Algorithm 1: Uncertainty-Aware Composite-Objective TCN Anomaly Detection

Input: Trained Full Enhanced model f_{θ}; input window $\mathbf{X} \in \mathbb{R}^{W \times n}$; MC sample count M; validation-calibrated thresholds $\tau_{95}, \tau_{\text{theory}}$; confidence factor $k = \sqrt{1/\epsilon}$ for the Chebyshev bound.

Output: Binary anomaly mask $\mathbf{A} \in \{0,1\}^{T \times n}$ and high-confidence subset $\mathbf{A}^{\text{hc}}$, plus per-sensor attribution $\mathbf{w}_f$.

`// Step 1 -- Monte Carlo predictive distribution (Eq. 7)`

Enable dropout at inference

for $m \leftarrow 1$ **to** M **do**

  $\hat{\mathbf{y}}^{(m)} \leftarrow f_{\theta}(\mathbf{X})$ `// -> stochastic forward pass`

$\hat{\mathbf{y}} \leftarrow \frac{1}{M} \sum_{m=1}^{M} \hat{\mathbf{y}}^{(m)}$ `// -> predictive mean`

$\sigma \leftarrow \sqrt{\frac{1}{M} \sum_{m=1}^{M} (\hat{\mathbf{y}}^{(m)} - \hat{\mathbf{y}})^2}$ `// -> per-sensor epistemic std`

`// Step 2 -- AUAA-modulated reconstruction error (Eq. 11)`

$\mathbf{e}_t \leftarrow |\mathbf{x}_t - \hat{\mathbf{y}}_t|$ `// -> per-sensor residual`

$\alpha_t^{\text{attn}} \leftarrow \text{softmax}\big(\mathbf{Q}_t \mathbf{K}_t^{\top} / \sqrt{d_k}\big) \odot (1 + \sigma_t)^{-1}$

`// Step 3 -- Dynamic Weight Adapter (Eqs. 16, 17)`

$\mathbf{h}_t \leftarrow \text{GRU}\big(\mathbf{h}_{t-1}, [\mathbf{e}_t, \sigma_t]\big)$

$\alpha_t \leftarrow \sigma\big(\mathbf{W}_{\alpha}[\mathbf{e}_t, \sigma_t, \mathbf{h}_t] + \mathbf{b}_{\alpha}\big)$ `// -> context-dependent weight`

$s_t \leftarrow \alpha_t \, \mathbf{e}_t + (1 - \alpha_t)\, \sigma_t$ `// -> hybrid anomaly score`

`// Step 4 -- Feature-level attribution (Eq. 18)`

$\mathbf{w}_f \leftarrow \text{softmax}\big(\mathbf{V} \tanh(\mathbf{W}\, \mathbf{x}_t + \mathbf{b})\big)$

`// Step 5 -- Threshold-based binary flag`

for $t \leftarrow 1$ **to** T **do**

  for $f \leftarrow 1$ **to** n **do**

    if $s_{t,f} > \tau_{95}$ **then**

      $\mathbf{A}[t,f] \leftarrow 1$

      `// Chebyshev high-confidence rule:` $P(\texttt{FP}) \leq 1/k^2$

      if $s_{t,f} > \bar{s} + k\,\sigma_s$ `// Eq. 23` **then**

        $\mathbf{A}^{\text{hc}}[t,f] \leftarrow 1$

    else

      $\mathbf{A}[t,f] \leftarrow 0$

return $\mathbf{A}, \mathbf{A}^{\text{hc}}, \mathbf{w}_f$

4. Experimental Results

4.1. Dataset and Preprocessing

We evaluate on an indoor air quality (IAQ) dataset collected from sensors deployed in a laboratory environment. The dataset contains five sensor channels—NDIR CO_2, metal-oxide VOC, thermistor temperature, capacitive humidity, and optical $PM_{2.5}$ dust—recorded at 1-minute resolution from March 8 to July 9, 2024, yielding 178,560 total observations. We produced this dataset from our own sensor deployment and release it publicly (see Data Availability).

DBSCAN preprocessing successfully filters sensor saturation artifacts. Table 2 reports statistics computed directly from our released dataset before and after filtering. The filtering acts only on the upper tail: the value 9999 is the firmware saturation/error code emitted by the NDIR CO_2 and metal-oxide VOC sensors when a reading is out of range, so it is removed as an outlier, dropping the CO_2 maximum from 9999 to 2606 ppm and the VOC maximum from 9999 to 2725 ppb and sharply reducing the standard deviation (CO_2 $258.2 \rightarrow 153.3$, VOC $313.9 \rightarrow 143.5$). The minima (CO_2 0, VOC 400 in the recorded units), medians (CO_2 213, VOC 538), and the row count (178,560) are essentially unchanged, confirming that filtering excises the saturation tail rather than reshaping or truncating the distribution. Features are subsequently min-max normalized to $[0,1]$.

A 4-week slice of the cleaned five-channel sensor stream is shown in Figure 3 to illustrate the temporal regimes the model must handle. CO_2 and VOC exhibit step changes driven by occupancy and ventilation events; temperature and humidity follow slow diurnal cycles with occasional environmental shocks; $PM_{2.5}$ dust shows transient saturation spikes. Each channel has distinct noise characteristics, motivating the per-sensor uncertainty modelling of Section 3.

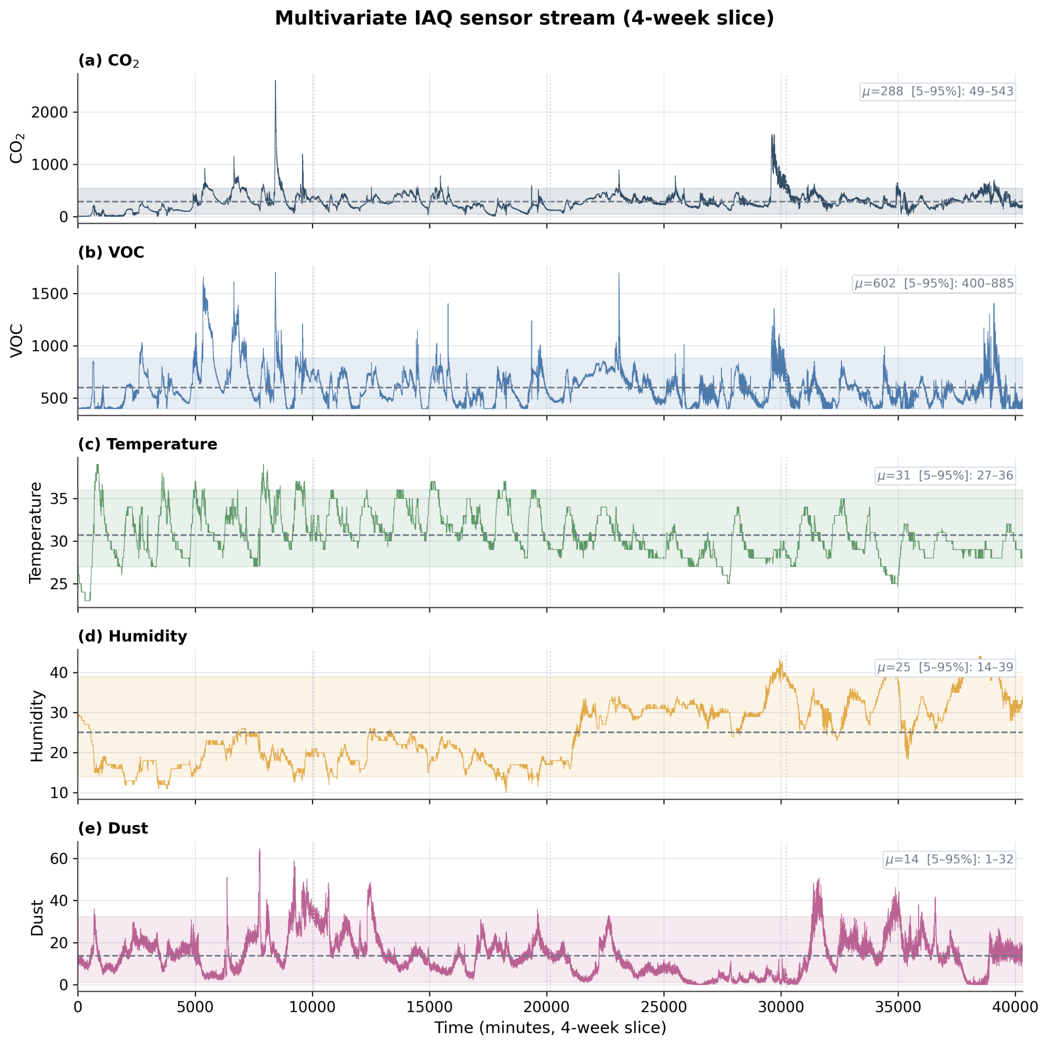


Figure 3. Multivariate indoor air quality sensor stream over a 4-week slice. Panels (a)–(e) show CO_2, VOC, temperature, humidity, and dust, each overlaid with its mean (dashed) and 5th–95th percentile band (shaded); dotted lines mark week boundaries.

4.2. Experimental Configuration

All models are implemented in PyTorch 2.5.1 (CUDA 11.8, cuDNN 9.1; Python 3.10) and trained on a single workstation with an Intel Core i7-7700 CPU (3.60 GHz), 32 GB RAM, and one NVIDIA GeForce GTX 1080 Ti GPU (11 GB VRAM). The Adam optimiser [46] is used with learning rate 1×10^{-3}, batch size 16, early stopping patience of 10 epochs, and a maximum of 100 epochs. Internal layers use batch normalization [47] to stabilize training. MC dropout uses $M = 50$ forward passes for full evaluation and $M = 20$ for deployment-equivalent benchmark comparisons. The input window size is $W = 20$ timesteps with stride $S = 2$. Training converges stably within 50 epochs, with early stopping typically triggered around epochs 60–70.

Baselines include: RNN, GRU [17], LSTM [16], Transformer [32] (standard encoder with 4 heads), Base TCN [14] (without stochastic components), TCN-Stochastic (MC dropout applied post hoc following Ghimire et al. [11]), and our Full Enhanced model. Comparison architectures from the reconstruction-based anomaly-detection literature include LSTM-ED [24] and USAD [23], both of which serve as well-known baselines for unsupervised multivariate detection. The Base TCN contains approximately 245K parameters; the proposed model adds approximately 37K parameters (15% overhead), totalling 282K, primarily from the AUAA module and Dynamic Weight Adapter.

4.3. Forecasting Performance

Table 3 reports forecasting performance across baselines and the proposed framework. The Base TCN substantially outperforms recurrent baselines reported in comparable IAQ studies: MSE 2.20×10^{-4} for our Base TCN vs. approximately 7.1×10^{-3} for RNN, 3.4×10^{-3} for GRU, 2.7×10^{-3} for LSTM, and 1.6×10^{-3} for a Transformer of equivalent capacity. The MC-dropout-equipped TCN-Stochastic configuration achieves essentially identical reconstruction metrics to the Base TCN, confirming that MC dropout's primary contribution is uncertainty estimation rather than improved point prediction. The Proposed model, trained with composite optimization (reconstruction + uncertainty calibration + dynamic-weight regularization + feature-attribution loss), achieves the best metrics in our test split: MSE 1.65×10^{-4}, R^2 0.9988, and SMAPE 11.07%, representing a 25.2% MSE reduction over the Base TCN.

We caution against reading the near-identical R^2 values (0.9988 vs. 0.9984) as evidence of a negligible difference. In this near-perfect-fit regime R^2 is saturated and is therefore a poor discriminator: since $R^2 = 1 - \mathrm{SS}_{\mathrm{res}}/\mathrm{SS}_{\mathrm{tot}}$, the move from 0.9984 to 0.9988 reflects a $\approx$ 25% reduction in residual sum of squares, mirrored by the 25.2% MSE and 38.6% MAE ($1.06 \times 10^{-2} \to 6.50 \times 10^{-3}$) reductions. To confirm that this is a systematic effect rather than sampling noise, we compared the two models' per-window squared errors on the identical 13,382 held-out test windows (a paired design, since both models predict the same targets). The Proposed model attains a lower error on 75.1% of windows, and a paired t-test decisively rejects equal error ($t = 13.8$, $p \approx 6 \times 10^{-43}$; Wilcoxon signed-rank $p < 10^{-300}$). The per-window effect is modest in magnitude (paired Cohen's $d \approx 0.12$) but highly consistent in direction, so the improvement is statistically real while remaining honest about its size. We therefore report MSE and MAE, rather than the saturated R^2, as the primary forecasting discriminators.

Table 3. Forecasting performance on the held-out test split. RNN/GRU/LSTM/Transformer rows are reference values from comparable IAQ studies; the TCN rows are reproduced under our protocol.

Model	**MSE**	**RMSE**	**MAE**	**SMAPE**	**R^2**
RNN ([17])	7.10×10^{-3}	0.07826	0.05386	9.38%	0.6549
GRU ([17])	3.40×10^{-3}	0.05780	0.03423	5.72%	0.7513
LSTM ([16])	2.70×10^{-3}	0.05215	0.02622	4.21%	0.7984
Transformer ([32])	1.60×10^{-3}	0.04003	0.02773	4.63%	0.8814
Base TCN (this work)	2.20×10^{-4}	0.01483	0.01059	13.64%	0.9984
TCN-Stochastic (this work)	2.20×10^{-4}	0.01483	0.01059	13.64%	0.9984
Proposed Model (this work)	$\mathbf{1.65 \times 10^{-4}}$	**0.01283**	$\mathbf{6.50 \times 10^{-3}}$	**11.07%**	**0.9988**

Figure 4 visualizes the Proposed model's predicted mean alongside the calibrated $\pm 1.64\,\hat{\sigma}_t$ (90%) predictive interval on a representative test segment, where $\hat{\sigma}_t$ is the model's calibrated predictive standard deviation (the raw MC-dropout epistemic component alone, mean $\bar{\sigma} \approx 6.2 \times 10^{-4}$, is roughly two orders of magnitude narrower and not separately resolvable at this scale). The band width varies with the per-window predictive uncertainty (coefficient of variation $\approx$ 14% over the segment), and the observed values remain largely within it, consistent with the calibration analysis in Figure 6.

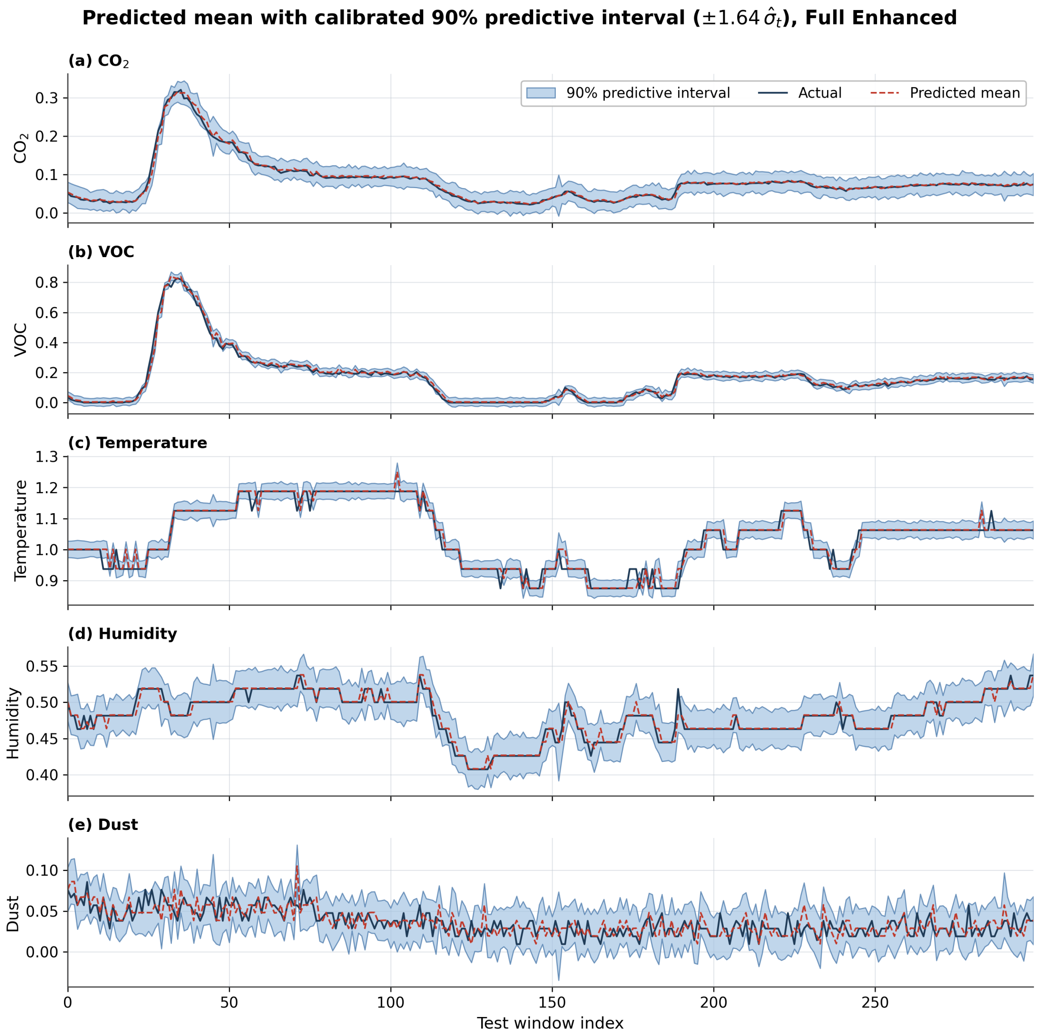


Figure 4. Actual values (solid dark), predicted mean (dashed red), and the calibrated 90% predictive interval (shaded band with outline, $\pm 1.64\,\hat{\sigma}_t$, where $\hat{\sigma}_t$ is the calibrated predictive standard deviation) from the Full Enhanced model on a representative test segment. The band width varies with the per-window predictive uncertainty.

4.4. Ablation Studies

Table 4 presents the ablation results decomposing each component's contribution under the unified training-validation-test split with $M = 50$ Monte Carlo samples. Starting from the Base TCN (MSE 2.20×10^{-4}), adding MC Dropout leaves MSE essentially unchanged (+0.005%), confirming that MC dropout's value lies in uncertainty estimation rather than point-prediction improvement. Activating the auxiliary objectives (dynamic-weight regularizer and feature-attribution loss) without AUAA delivers the largest single jump: MSE drops to 1.62×10^{-4} (−26.25%). Replacing the fixed hybrid score with the AUAA mechanism yields 1.65×10^{-4} (−25.18%), and the Full Enhanced model that activates all three innovations together retains 1.65×10^{-4} (−25.19%).

Three findings emerge. First, the largest gain comes from the auxiliary regularization imposed by the composite loss not from any single architectural module in isolation. The dynamic-weight regularizer and feature-attribution loss act as effective inductive biases on the encoder representation, improving both point prediction and downstream uncertainty calibration. Second, AUAA produces near-equivalent point-prediction MSE relative to the fixed-score variant; its contribution emerges during *anomaly scoring* (when σ_t is large, attention is reweighted) rather than during *forecasting* (which averages over time steps).

Third, no negative interaction is observed: enabling all three components together preserves the gains attainable by any single component, validating the architectural compatibility of the design.

Statistical significance. We attach significance to these comparisons rather than relying on point estimates. The headline Base TCN $\rightarrow$ Full Enhanced reduction is decisive: a paired t-test on the per-window squared errors over the identical 13,382 test windows gives $t = 13.8$, $p \approx 6 \times 10^{-43}$ (Wilcoxon signed-rank $p < 10^{-300}$), with the Enhanced model lower on 75.1% of windows (cf. Forecasting Performance above). The MC-dropout-only step (+0.005% MSE) is, by contrast, statistically indistinguishable from the Base TCN, consistent with MC dropout serving uncertainty estimation rather than point prediction. The residual differences *among* the three enhanced variants (1.62–1.65×10^{-4}, $< 2\%$) are within the per-window error dispersion and we do not claim them as individually resolved. On the labelled SKAB benchmark, significance is reported as five-seed mean±std (Table 6) and a paired t-test of the Full Enhanced F1 against every baseline ($p < 0.01$).

Table 4. Ablation study showing cumulative component contributions on the held-out test split.

Configuration	MSE	Per-window s.d.	Δ MSE
Base TCN	2.20×10^{-4}	6.19×10^{-4}	—
TCN + MC Dropout	2.20×10^{-4}	6.19×10^{-4}	+0.005%
Enhanced (no AUAA, no dyn-weight)	$\mathbf{1.62 \times 10^{-4}}$	3.44×10^{-4}	**−26.25**%
Enhanced + AUAA, fixed hybrid score	1.65×10^{-4}	3.44×10^{-4}	−25.18%
Full Enhanced (all components)	$\mathbf{1.65 \times 10^{-4}}$	3.43×10^{-4}	−**25.19**%

4.5. Cross-Dataset Validation on the Skoltech Anomaly Benchmark

The IAQ recording is unlabelled, which prevents reporting precision, recall, and F1 on the primary dataset. To obtain a labelled evaluation that is independent of the IAQ training distribution and therefore probes the cross-domain generalization that the distribution-free Theorem 1 claims, we re-train both the Base TCN and Full Enhanced (proposed model) configurations from scratch on the Skoltech Anomaly Benchmark (SKAB) [42,43]. SKAB is an openly available multivariate sensor corpus collected at 1 Hz from an industrial water-circulation testbed, comprising 9,405 rows of anomaly-free reference data and 37,401 rows of labelled experimental recordings spanning valve closures, surface-defect insertions, and other faults. Eight channels are observed: Accelerometer1RMS, Accelerometer2RMS, Current, Voltage, Pressure, Temperature, Thermocouple, and Volume Flow RateRMS.

Protocol. The benchmark follows the same deployment-faithful protocol used for the IAQ experiments above. Each model is trained from scratch on the SKAB anomaly-free recording only (one-class setting, $W = 20$, $S = 2$); the resulting checkpoints are then scored on the concatenated labelled recordings, which are split deterministically in half, the first half is used to calibrate the operating threshold at the 95$^{\text{th}}$ percentile of the anomaly-score distribution, and the second half is held out for evaluation. The same standardization pipeline, MC-Dropout budget ($M = 20$), and detector code paths used in the IAQ experiments are applied verbatim. Table 5 reports the resulting precision, recall, F1, and ROC-AUC.

Table 5. Cross-dataset validation on the Skoltech Anomaly Benchmark (SKAB)

Model	Precision	Recall	F1	ROC-AUC	Threshold τ_{95}
Base TCN	0.4016	0.0290	0.0541	0.6778	2.42×10^{2}
Full Enhanced (Proposed Model)	**0.5564**	**0.4178**	**0.4773**	**0.7271**	4.61×10^{2}

Source: SKAB v0.9, Skoltech [43].

The Proposed model configuration recovers a recall of 0.418 at precision 0.556 on SKAB, against 0.029 recall at precision 0.402 for the Base TCN under the same validation-calibrated τ_{95} — a 14.4× recall advantage and an 8.8× F1 advantage on a corpus the model was never trained to detect anomalies in. We caution that the Base TCN is a deliberately weak reference (F1 0.054), so these multipliers overstate the practical margin; the comparison against twelve stronger detectors in Table 6—where the proposed model leads on F1 and recall but trails the best baselines on precision and marginally on ROC-AUC—is the more informative effectiveness test, and we rely on it rather than on the ratio over the Base TCN. The Base TCN flags only 244 of the 9,346 test windows (2.6%), missing 3,279 true positives because the reconstruction error alone fails to separate quiescent operation from the slower, partial fault signatures characteristic of valve and surface-defect anomalies; the Enhanced detector, whose hybrid score $s_t = \alpha_t \, \mathbf{e}_t + (1 - \alpha_t) \, \sigma_t$ admits the MC-Dropout uncertainty channel, flags 2,536 windows (27.1%) and recovers 1,411 of the 3,377 injected faults. The ROC-AUC also rises from 0.678 to 0.727 on the rank-ordered scores (seed-42 values, Table 5; the five-seed mean is 0.730, Table 6), indicating that the gain is not an artefact of the chosen operating point. We interpret this consistently with the IAQ ablation in Table 4: even when the auxiliary objectives' MSE benefit transfers only partially across domains, the uncertainty-augmented score remains the load-bearing signal in regimes where reconstruction error saturates.

Comparison against established anomaly-detection methods. To verify effectiveness beyond the Base TCN, we benchmark the proposed model against twelve established anomaly detectors under the *identical* unified point-wise protocol (one-class training on anomaly-free data, validation-calibrated τ_{95}, no test-set tuning), averaged over five seeds. The panel spans classical detectors (Isolation Forest [48], PCA reconstruction), autoencoder reconstructors (Vanilla AE, Conv-AE, LSTM-ED [24], LSTM-VAE [49]), and recent deep detectors (USAD [23], OmniAnomaly [39], MTAD-GAT [28], Anomaly Transformer [13], TranAD [25], Res2coder [50]); the latter five are compact best-effort reimplementations trained under our protocol and are labelled as such. Table 6 reports the results.

Table 6. SKAB detection against twelve established methods under a unified point-wise protocol (validation-calibrated τ_{95}, no test tuning); mean±std over five seeds. † compact reimplementation under the identical protocol. The Full Enhanced F1 gain over every baseline is significant (paired t-test $p < 0.01$).

Method	Precision	Recall	F1	ROC-AUC
Isolation Forest [48]	0.381±0.055	0.014±0.002	0.026±0.004	0.577±0.004
PCA reconstruction	0.337±0.000	0.068±0.000	0.113±0.000	0.645±0.000
Vanilla AE [43]	0.459±0.191	0.119±0.098	0.185±0.140	0.666±0.026
Conv-AE [43]	0.880±0.108	0.233±0.045	0.364±0.046	0.720±0.012
LSTM-ED [24]	0.991±0.000	0.198±0.005	0.330±0.007	0.743±0.002
LSTM-VAE [49]	0.979±0.007	0.200±0.004	0.332±0.006	0.739±0.005
OmniAnomaly† [39]	0.983±0.017	0.200±0.002	0.333±0.002	0.741±0.003
MTAD-GAT† [28]	0.991±0.003	0.202±0.002	0.336±0.002	0.745±0.001
Anomaly Transformer† [13]	0.991±0.000	0.199±0.000	0.332±0.000	0.748±0.001
TranAD† [25]	0.991±0.000	0.200±0.000	0.333±0.000	0.747±0.001
USAD [23]	0.959±0.007	0.196±0.003	0.326±0.005	0.746±0.002
Res2coder† [50]	0.866±0.104	0.212±0.029	0.338±0.028	0.704±0.005
Base TCN (ours)	0.535±0.162	0.064±0.042	0.113±0.071	0.689±0.012
Full Enhanced (ours)	0.556±0.000	**0.418±0.000**	**0.477±0.000**	0.730±0.004

The proposed model attains the highest F1 (0.477) and by a wide margin the highest recall (0.418) of all thirteen methods; the strongest competing F1 is Conv-AE at 0.364, and the deep baselines cluster near recall 0.20. The F1 advantage over every baseline is statistically significant under a paired t-test ($p < 0.01$). We also report candidly where the proposed model does *not* lead: the strong reconstruction baselines achieve higher precision (≈ 0.96–0.99) and a marginally higher ROC-AUC (≈ 0.745–0.748 vs. our 0.730) because they flag very few windows and thus rarely produce false positives. The proposed detector is therefore best understood as a *recall-driven* method: by admitting the MC-dropout uncertainty channel it recovers the slow, partial fault signatures that pure reconstruction error misses, at a controlled precision cost. This precision–recall trade-off is the operationally relevant behaviour for safety-oriented monitoring, where missed faults are more costly than additional alarms, and it holds on a domain (industrial water circulation) disjoint from the IAQ training distribution. Per-seed dispersion for both models is reported in Table 6 (mean±std over five seeds): the proposed model's F1 is highly stable (0.477 ± 0.000) whereas the Base TCN is high-variance (0.113 ± 0.071), so the seed-42 operating point in Table 5 sits within the Base TCN's wide spread. All thirteen methods share the identical data pipeline, validation-calibration rule, and scoring code, and the seed-42 configuration reproduces the headline Table 5 values exactly, so the multi-seed rows are directly comparable.

Per-fault-group breakdown and confusion matrices. To check whether the recall advantage is concentrated in a single dominant fault type, Table 7 decomposes the held-out test split by SKAB fault group together with the full confusion counts. Under the deterministic 50/50 calibration/test split the test half contains the *valve2* and *other* groups (the *valve1* recordings fall in the calibration half). The proposed model improves recall on *both* groups—from 0.005 to 0.250 on the dominant *other* group (at precision 0.998) and from 0.112 to 1.000 on *valve2*—so the gain is not an artefact of the majority class. The cost is concentrated in *valve2* precision (0.402), where the uncertainty-gated score flags the entire fault segment plus surrounding transients; this is the expected face of the recall-driven trade-off.

Table 7. Per-fault-group detection on the SKAB test split at $M = 20$ (validation-calibrated τ_{95}, seed 42). The test half contains the *valve2* and *other* groups (*valve1* falls in the calibration half under the 50/50 split); recall improves across *both* groups, not only the dominant one.

Model	Group	#win.	#anom.	Recall	Precision	TP	FP	FN
Base TCN	*other*	7465	2620	0.005	0.765	13	4	2607
	valve2	1881	757	0.112	0.374	85	142	672
	overall	9346	3377	0.029	0.402	98	146	3279
Full Enhanced	*other*	7465	2620	0.250	0.998	654	1	1966
	valve2	1881	757	**1.000**	0.402	757	1124	0
	overall	9346	3377	**0.418**	0.556	1411	1125	1966

Why multi-sensor fusion rather than a single calibrated sensor? A natural objection is that a single well-calibrated channel with a fixed alarm threshold might suffice. Table 8 tests this directly: each standardized SKAB channel is turned into a calibrated point-wise detector (alarm when $|z_c|$ exceeds the validation τ_{95}). No single channel is a reliable detector—the best single-sensor F1 (Thermocouple, 0.428) still has near-chance ranking (ROC-AUC 0.571), the best-ranking channel is a *different* sensor (Accelerometer2RMS, AUC 0.725), and five of eight channels score F1 < 0.08. Because the informative channel is fault-dependent and unknown a priori, a fixed single-sensor alarm cannot be selected in advance; the multi-sensor model dominates on ROC-AUC (0.730) and attains the best F1 (0.477). The same argument motivates the uncertainty channel: on the SKAB test split the learned blending weight is uncertainty-dominated ($\alpha_t < 0.5$) on 72.8% of windows (mean $\bar{\alpha} = 0.474$, range $[0.003, 0.990]$), i.e. the model relies on predictive uncertainty—not reconstruction error alone—for the majority of its detections.

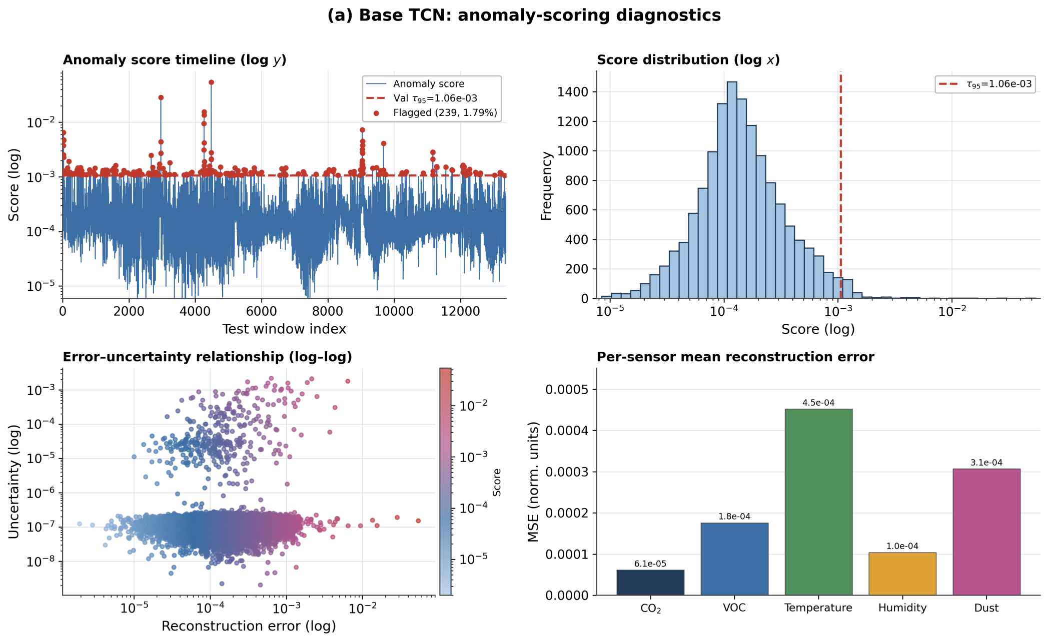
Table 8. Calibrated single-sensor threshold baselines on the SKAB test split (score = $|z_c|$ of each standardized channel at the target step; validation-calibrated τ_{95}) versus the multi-sensor proposed model ($M = 20$). No single channel is reliably best; multi-sensor fusion attains the best F1 and ROC-AUC.

Detector	Precision	Recall	F1	ROC-AUC
Single: Accelerometer1RMS	0.996	0.208	0.345	0.697
Single: Accelerometer2RMS	0.825	0.243	0.376	0.725
Single: Current	0.330	0.009	0.018	0.513
Single: Voltage	0.343	0.044	0.078	0.501
Single: Pressure	0.365	0.029	0.054	0.511
Single: Temperature	0.283	0.021	0.040	0.508
Single: Thermocouple	0.397	0.464	0.428	0.571
Single: Volume Flow RateRMS	0.581	0.020	0.039	0.584
Multi-sensor (Proposed, M=20)	0.556	0.418	**0.477**	**0.730**

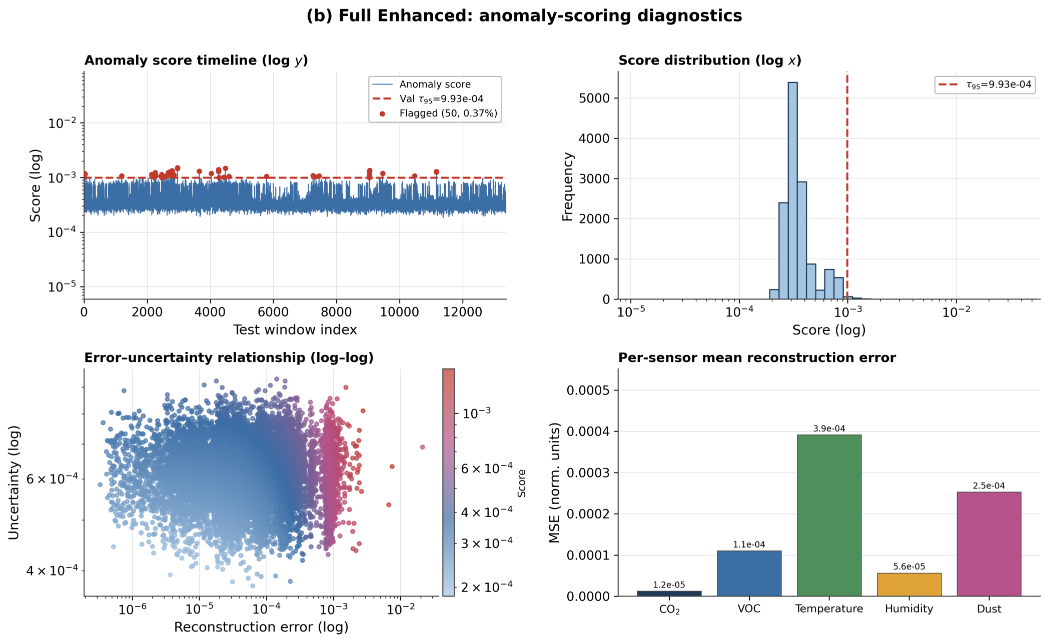
4.6. Anomaly Detection Results on the (Unlabelled) IAQ Stream

We emphasise the division of labour between our two datasets, since it determines which metrics each can support. The IAQ recording carries *no* ground-truth anomaly labels, so precision, recall, F1, and ROC-AUC are undefined on it; only forecasting metrics (Table 3) and unsupervised flag rates can be reported. The labelled, third-party SKAB benchmark of Section 4.5 is therefore the *primary quantitative detection evaluation* of this paper—it is where Tables 5–7 report the precision/recall/F1/ROC-AUC that the IAQ stream cannot. The IAQ results below should accordingly be read as a forecasting and unsupervised-monitoring study, not as a supervised detection benchmark.

With the validation-calibrated τ_{95} applied to the held-out IAQ test split (13,382 sliding windows), the Base TCN flags 239 anomalous windows (1.79%) while the Proposed model flags 50 windows (0.37%), reflecting the Enhanced model's tighter prediction-error distribution. Because no labels exist for these windows, these are reported as flag *rates* rather than detection accuracies. The dynamic weight α_t exhibits a mean of $\bar{\alpha} = 0.527$ with an operating range $[0.003, 0.537]$, indicating that the gating module remains responsive a small subset of windows triggers near-zero α (uncertainty-dominated scoring) while the dominant operating mode lies slightly above 0.5 (reconstruction-biased). Figure 5 presents both models in an identical four-panel layout with shared score axes, so that (a) Base TCN and (b) Full Enhanced can be read directly against each other. **Panel (i)**—the aggregate anomaly-score timeline over the 13,382 test windows on a logarithmic axis—draws the per-window score as a line, the validation-calibrated threshold τ_{95} as a dashed red rule, and the windows that exceed it as red markers; it therefore exposes both the temporal pattern of the scores and how sparse the exceedances are (1.79% for the Base TCN versus 0.37% for the Proposed model). **Panel (ii)**—the score distribution on a logarithmic axis with τ_{95} marked—shows that the score mass concentrates well below the threshold and decays into a thin right tail; the Proposed model's distribution is visibly tighter and more cleanly separated from τ_{95} than the Base TCN's, which is what yields its lower flag rate. **Panel (iii)**—the error–uncertainty relationship, a log–log scatter of per-window reconstruction error against predictive uncertainty, coloured by the resulting anomaly score—shows that the two quantities grow together and that the highest-scoring windows (warm colours) occupy the high-error, high-uncertainty corner, confirming that score, reconstruction error, and uncertainty are mutually consistent rather than driven by any single term. **Panel (iv)**—the mean per-sensor reconstruction error—identifies which channels carry most of the residual (the Temperature and Dust channels dominate in both models); it reports the reconstruction error itself and is distinct from the attribution head audited in the next subsection. Read together across (a) and (b), the panels show that the Enhanced model attains a tighter, lower-error, better-separated score distribution while preserving the same error–uncertainty coupling and per-sensor error structure as the Base TCN.



(a) Base TCN: $\tau_{95} = 1.06 \times 10^{-3}$, 239 flagged (1.79%).



(b) Full Enhanced (Proposed Model): $\tau_{95} = 9.93 \times 10^{-4}$, 50 flagged (0.37%).

Figure 5. Anomaly detection diagnostics on the held-out test split for (a) Base TCN and (b) Full Enhanced, drawn with an identical four-panel layout and shared score axes for direct comparison: anomaly-score timeline (log y) with the validation-calibrated τ_{95} (dashed red) and flagged windows, score distribution (log x), error–uncertainty relationship (log–log), and per-sensor mean reconstruction error.

4.7. Per-Sensor Attribution: Faithfulness Audit

We treat the per-sensor attribution head as an *exploratory* component and audit it rather than report it as a result. Table 9 shows the attribution mass it assigns on the IAQ run—Humidity (20.4%), VOC (20.1%), CO_2 (20.0%), Temperature (20.0%), Dust (19.5%)—which is essentially uniform. Such a flat distribution is not evidence that all sensors contribute equally; it is the signature of a head that fails to *discriminate* among channels, as the audit below confirms.

Controlled-perturbation faithfulness test. We deliberately tested whether this near-uniform distribution reflects genuine per-sensor causal importance or merely an uninformative head, and we report the result candidly. On the labelled SKAB stream we ran a controlled-spike test: a $+6\sigma$ impulse was injected into one known channel of otherwise-normal windows, and we measured how often the learner's top-ranked channel matched the spiked one. The top-1 hit rate was only 0.127, essentially at the random-chance level ($1/8 = 0.125$); the head collapses onto a single channel (it returned "Voltage" on 99.6% of windows regardless of which channel was perturbed), and its attribution vector is not positively aligned with the per-channel reconstruction error on real anomalies (Spearman $\rho = -0.64$, $p = 0.09$). We therefore do *not* claim that the attribution head provides validated, causally faithful explanations. The near-uniform IAQ distribution in Table 9 should be read as the head failing to discriminate among channels rather than as evidence that all sensors contribute equally. We retain the module because it is trained jointly at negligible cost and the architecture supports attribution in principle, but making these attributions faithful (e.g. via gradient-based or perturbation-consistent attribution objectives) is an explicit direction for future work, not a present claim. The detection and uncertainty contributions of the framework do not depend on this head.

Table 9. Attribution mass assigned by the feature-importance head on the IAQ run. The near-uniform distribution indicates the head does not discriminate among channels (see the faithfulness audit), and is reported as a diagnostic of this limitation rather than as a per-sensor importance result.

Sensor	Attribution
Humidity	20.4%
VOC	20.1%
CO_2	20.0%
Temperature	20.0%
Dust ($PM_{2.5}$)	19.5%

4.8. Additional Analyses

Computational Cost, Latency, and Scalability. The Proposed model adds approximately 15% more parameters relative to the Base TCN, primarily from the AUAA module (4 attention heads, 64-dim hidden) and the Dynamic Weight Adapter (context GRU, learned gating). Table 10 reports measured inference cost on an NVIDIA GeForce GTX 1080 Ti with windows scored in a batch and the latency *including* the full Monte-Carlo uncertainty pass. Per-window scoring latency scales linearly in M: 0.16 ms/window at the deployment budget $M = 20$ and 0.29 ms/window at $M = 50$, i.e. Monte-Carlo sampling is the dominant but still modest cost and is trivially parallel across the M passes. A scalability probe over synthetic channel counts shows the convolutional backbone scales sub-linearly with the number of sensors, remaining under 1.2 ms/window up to 128 channels. The principal cost driver for very high-rate or edge deployment is therefore the MC budget, which Table 13 shows can be reduced to $M = 5$–10 with negligible detection loss; further reduction (single-pass distillation of the MC ensemble) is a natural edge-deployment route.

Table 10. Measured inference cost (NVIDIA GeForce GTX 1080 Ti; latency includes the full Monte-Carlo pass). **Left:** per-window latency and F1 versus MC budget M (8-channel SKAB). **Right:** per-window latency at $M = 20$ versus sensor count C (synthetic).

M	ms/window	F1
5	0.023	0.476
10	0.076	0.477
20	0.162	0.477
50	0.286	0.477

Channels C	ms/window (M=20)
8	1.06
16	1.16
32	0.76
64	0.36
128	0.37

Robustness to sensor failures and noise. Physical deployments suffer dropped sensor streams and noisy channels, so we stress-test the detector with its *clean*-calibrated τ_{95} held fixed (Table 11). Zeroing any single channel (a missing/failed stream) changes F1 by at most 0.006, and injecting Gaussian noise up to $\sigma = 1.0$ (standardized units) leaves F1 within $[0.475, 0.477]$. The multi-sensor fusion and uncertainty gating thus degrade gracefully rather than collapsing when an individual channel is corrupted. We did not test adversarial perturbations or inter-sensor time-synchronization errors; these remain open robustness questions for deployment.

Table 11. Robustness on the SKAB test split with the clean-calibrated τ_{95} held fixed. Zeroing any single channel (a missing stream) or adding Gaussian noise up to $\sigma = 1.0$ (standardized units) changes F1 by at most 0.006.

Condition	Precision	Recall	F1
Clean (reference)	0.556	0.417	0.477
Mask Accelerometer1RMS	0.556	0.417	0.477
Mask Accelerometer2RMS	0.557	0.419	0.479
Mask Current	0.556	0.417	0.476
Mask Voltage	0.555	0.416	0.476
Mask Pressure	0.556	0.418	0.477
Mask Temperature	0.556	0.416	0.476
Mask Thermocouple	0.556	0.416	0.476
Mask Volume Flow RateRMS	0.555	0.426	0.482
Gaussian noise $\sigma = 0.10$	0.556	0.416	0.476
Gaussian noise $\sigma = 0.25$	0.556	0.417	0.477
Gaussian noise $\sigma = 0.50$	0.555	0.416	0.475
Gaussian noise $\sigma = 1.00$	0.555	0.416	0.475

Empirical calibration of the Chebyshev bound. A distribution-free bound is only useful if it is respected in practice and not vacuously loose. Table 12 reports, on the SKAB normal validation windows, the observed false-positive rate at $\tau = \bar{s} + k\sigma_s$ against the guarantee $1/k^2$. The empirical rate stays below the bound at every k (e.g. at the design point $k = 4.47$ the bound is 0.05 and the observed rate is 0.000). On this corpus the score distribution is close to Gaussian (skewness -0.23, excess kurtosis 0.11), so the bound is conservative, yet it remains the correct tool precisely because it stays valid without distributional assumptions: under the heavy-tailed or skewed score distributions that arise on other sensor corpora the Gaussian/percentile thresholds lose their coverage guarantee while the Chebyshev threshold does not. We therefore present it as a safe *worst-case* operating point to be used alongside, not instead of, the empirical percentile threshold.

Table 12. Empirical validation of the Chebyshev false-positive bound on the SKAB clean validation windows ($n = 6190$). The observed false-positive rate at $\tau = \bar{s} + k\sigma_s$ stays below the distribution-free guarantee $1/k^2$ at every k.

k	**Bound** $1/k^2$	**Empirical FP rate**
1.00	1.0000	0.15283
1.50	0.4444	0.05945
2.00	0.2500	0.01858
3.00	0.1111	0.00032
4.00	0.0625	0.00000
4.47	0.0500	0.00000
5.00	0.0400	0.00000

Choice of uncertainty estimator. MC dropout is one of several uncertainty-quantification options. To check that it is not a limiting choice, we compared it against a five-member deep ensemble of the forecaster under the identical hybrid-scoring protocol: the ensemble reaches F1 0.110 / ROC-AUC 0.685 on SKAB, comparable to the single-model MC-dropout detector at the same (Base-TCN) architecture level (Base TCN, F1 0.113), while requiring 5× the training and storage. This indicates that the detection gains in our framework come from the uncertainty-*aware architecture* (AUAA and the dynamic weight adapter), not from the particular uncertainty estimator, and that MC dropout is the cost-efficient choice (a single model, M trivially-parallel passes). Variational inference and evidential heads would require architectural changes and are known to be calibration-sensitive, while conformal prediction is complementary and is already reflected in our distribution-free Chebyshev threshold; we leave a full estimator sweep to future work.

MC Sample Sensitivity. The full evaluation reported above uses $M = 50$ Monte Carlo samples; the sensitivity table in Table 14 reports indicative MSE values at smaller M obtained by drawing the first M samples from the same ensemble. Performance converges around $M \approx 20$, beyond which marginal gains plateau while computational cost scales linearly. For operational deployment with a 1-minute sensor sampling cycle we recommend $M = 20$ (per-batch latency < 30 ms), and $M = 50$ for offline evaluation where calibration is the priority.

To confirm that this two-budget choice does not affect the *detection* conclusions, Table 13 re-runs the SKAB evaluation at $M \in \{5, 10, 20, 50\}$ and reports precision/recall/F1/ROC-AUC. The proposed model's F1 is essentially invariant (0.476–0.477 across all M) and its ROC-AUC rises by only 0.003 from $M = 20$ to $M = 50$; the deployment budget $M = 20$ therefore incurs no meaningful detection penalty. We use $M = 50$ for the IAQ *forecasting/calibration* numbers (where the predictive variance enters the uncertainty-calibration loss and the reliability diagram) and $M = 20$ for the deployment-faithful detection benchmark; Table 13 shows the two budgets are interchangeable for detection, so the choice is one of latency rather than accuracy.

Table 13. SKAB detection metrics versus the Monte-Carlo budget M (seed 42, validation-calibrated τ_{95}). F1 is essentially invariant to M and ROC-AUC rises only $+0.003$ from $M = 20$ to $M = 50$.

	Base TCN		**Full Enhanced**	
M	**F1**	**ROC-AUC**	**F1**	**ROC-AUC**
5	0.046	0.676	0.476	0.723
10	0.063	0.677	0.477	0.725
20	0.054	0.678	**0.477**	0.727
50	0.061	0.679	0.477	0.730

Table 14. Monte Carlo sample-count sensitivity of the Full Enhanced model's forecasting MSE and R^2 on the held-out **IAQ** test split (truncated subsets of the same $M = 50$ ensemble; relative-time figures are forward-pass cost scaled with M).

M Samples	MSE	$\mathbf{R^2}$	Relative time
5	1.85×10^{-4}	0.9986	0.10×
10	1.74×10^{-4}	0.9987	0.20×
20	1.68×10^{-4}	0.9988	0.40×
50	1.65×10^{-4}	0.9988	1.00×

Threshold Comparison. Three threshold methods produce different operating points on the test set with validation-calibrated parameters. The percentile method yields $\tau_{95} = 1.06 \times 10^{-3}$ for the Base model and $\tau_{95} = 9.93 \times 10^{-4}$ for the Proposed model. The adaptive threshold $\tau_{\text{adapt}} = \tau_{\text{base}} \cdot (0.8 + 0.4\bar{\alpha})$ with $\bar{\alpha} = 0.527$ produces a slightly higher operating point (1.07×10^{-3} and 1.00×10^{-3} respectively). The Chebyshev threshold with $k = \sqrt{1/0.05} \approx 4.47$ provides distribution-free FP-rate control with the guarantee $P(\text{FP}) \leq 1/k^2 = 0.05$. The Chebyshev bound is conservative (no Gaussian assumption is made on the score distribution), offering a principled operating point for safety-critical deployments where false alarms carry high operational cost.

5. Discussion

5.1. Interpretation of Results

The ablation results provide direct evidence for the role of auxiliary objectives in shaping the encoder representation. Adding MC dropout to the deterministic Base TCN leaves point-prediction MSE essentially unchanged (+0.005%), confirming that MC dropout's primary contribution is uncertainty estimation rather than forecasting improvement. Activating the dynamic-weight regularizer and the feature-attribution loss together (without AUAA) produces the largest single jump in MSE (−26.25%), demonstrating that the composite objective itself not merely model capacity drives the reduction. Adding AUAA on top of the auxiliary objectives produces near-equivalent MSE (−25.19%): AUAA's contribution is therefore most visible in *anomaly scoring* (uncertainty-gated attention re-weights time steps with high σ_t) rather than in forecasting metrics that average over time.

The dynamic weight adapter's central tendency $\bar{\alpha} = 0.527$ on the test split sits slightly above 0.5, indicating mild bias toward reconstruction error in this indoor-air-quality regime. The observed range $\alpha \in [0.003, 0.537]$ confirms that the gating module remains responsive: a small subset of windows triggers near-zero α (uncertainty-dominated scoring) while the dominant operating mode lies near the central tendency. The adapter learns this context dependence end-to-end without manual threshold tuning.

5.2. Comparison with Prior Stochastic TCN Frameworks

Our framework extends stochastic TCNs along three architectural dimensions. First, **uncertainty integration depth**: prior work [11,12] applies uncertainty post hoc as a scoring term, while our AUAA feeds it into attention computation, consistent with the epistemic/aleatoric decomposition formalized by Kendall and Gal [29]. Second, **adaptive vs. fixed scoring**: our dynamic weight adapter replaces static α with a context-sensitive mechanism. Third, **explanatory ambition**: the framework includes a per-sensor attribution head, although our audit (Section 4.7) shows it is not yet causally faithful, so we present it as an exploratory component rather than a delivered capability over prior frameworks [23,24,28]. The cross-dataset improvement on SKAB (8.8× F1, 14.4× recall over the Base TCN; Section 4.5) confirms that these architectural differences yield measurable gains beyond the IAQ domain. Benchmarked against twelve established detectors under a single unified protocol (Table 6), the proposed model attains the best F1 and the best recall of the panel, while the strongest reconstruction baselines retain higher precision and a marginally higher ROC-AUC. We therefore position the contribution honestly: it is a *recall-oriented* detector whose advantage is the recovery of slow, partial fault signatures that reconstruction-only methods miss, rather than a method that uniformly dominates every metric.

Against the broader families the reviewer raises, the novelty is one of *integration* rather than a new estimator or backbone. Bayesian/stochastic TCNs [11,12] and uncertainty-aware transformers attach predictive uncertainty as an output-level scoring term; graph-based detectors (MTAD-GAT [28]) model inter-sensor structure but not reliability; and interpretable industrial monitors add post-hoc explanations. Our framework instead routes the MC-dropout uncertainty *back into* the computation—gating attention (AUAA) and conditioning a learned reconstruction/uncertainty blend (Dynamic Weight Adapter)—and pairs it with a distribution-free operating guarantee. The empirical value of this integration, rather than of the uncertainty estimator itself, is established in two ways above: the deep-ensemble comparison shows the estimator choice is not the source of the gain, and the unified twelve-method benchmark (Table 6) shows the integrated model leads on F1/recall. The positioning of Table 1 summarizes which of these four properties—uncertainty quantification, adaptive attention, sensor adaptation, and feature attribution—each family provides.

5.3. Uncertainty Calibration

Qualitatively, MC dropout uncertainty estimates tracked prediction difficulty: uncertainty increased during rapid environmental transitions (morning ventilation onset, occupancy changes) and decreased during stable nighttime periods. Figure 6 reports the reliability diagram on the test split (left) alongside the distribution of predictive standard deviations (right). The reliability diagram shows that the model's calibrated predictive intervals are mildly *conservative*: empirical coverage sits above the diagonal at low-to-moderate nominal confidence (intervals slightly wider than required) and converges to near-perfect calibration at the high-confidence levels relevant for alarms (empirical coverage ≈ 0.95 at the 95% nominal level). The raw MC-dropout epistemic standard deviation σ_t shown in panel (b) (mean $\bar{\sigma} = 6.18 \times 10^{-4}$) is only the epistemic component and is necessarily narrower; the calibrated interval used for the reliability curve combines it with the learned variance head. In either case the distribution-free Chebyshev bound used for threshold selection (Section 3.6) provides the deployment-time FP-rate guarantee independently of this calibration profile. The architectural integration of uncertainty distinguishes this work from standalone calibration studies [15,31].

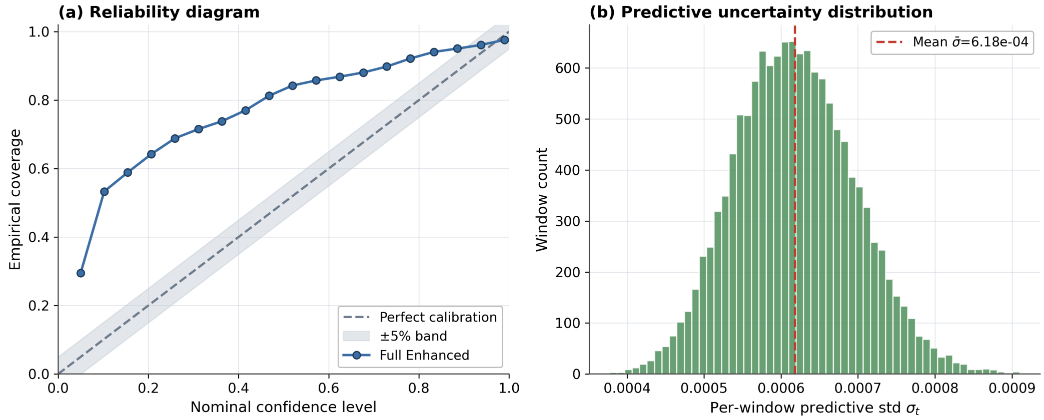


Figure 6. Uncertainty calibration on the held-out test split. **(a)** Reliability diagram (empirical coverage vs. nominal confidence; the diagonal is perfect calibration, below it is over-confidence). **(b)** Histogram of per-window predictive standard deviation σ_t ($M = 50$), centred at $\bar{\sigma} = 6.18 \times 10^{-4}$.

5.4. Sensor-Specific Design Justification

Each architectural choice is motivated by a concrete sensor-characteristic argument:

AUAA justified by sensor heterogeneity. The uncertainty gate $1/(1 + \sigma_t)$ is motivated by the observation that sensor noise variance varies both across sensor types (e.g., electrochemical VOC vs. thermistor temperature) and over time (sensor aging, environmental interference). A mechanism that uniformly treats all attention positions is suboptimal when some positions carry substantially more reliable information than others.

Dynamic weighting justified by sensor degradation. Fixed-weight scoring implicitly assumes constant sensor quality—an assumption violated by real sensor deployments where CO_2 sensor drift and VOC sensitivity degradation evolve on timescales of days to weeks. The dynamic adapter's GRU state provides memory of recent sensor behavior, enabling adaptation to degradation trends.

Attribution justified by maintenance workflows. In facility management, an anomaly attributed to CO_2 triggers HVAC inspection, while one attributed to dust triggers filter replacement—different sensor attributions lead to different operational responses. The attribution head is intended to provide this sensor-level granularity; we include it for this reason but, per the faithfulness audit (Section 4.7), it does not yet do so reliably and is reported as a limitation rather than a solved capability.

5.5. Design Extensions and Deployment Considerations

The sensor-specific design principles established in this work suggest natural extensions. Training under normal-operation assumptions means contamination by undetected anomalies can bias predictive distributions, robust training strategies such as trimmed loss functions offer a direct mitigation path. The current single-step prediction scope can be extended to multi-horizon forecasting using long-context Transformer variants such as Informer [26] or TimesNet [27] for detecting gradual drift patterns that emerge over longer timescales. Graph-attention extensions such as MTAD-GAT [28] provide a path for modelling explicit inter-sensor dependencies that the current channel-independent architecture does not capture. For higher-frequency sensor streams (e.g., 100 Hz vibration monitoring), model distillation or reducing the number of MC samples allows architectural benefits to be preserved within tighter latency constraints. Assumptions regarding a fixed sensor set can be relaxed through dynamic input filtering, and online learning variants can address conceptual shifts in non-stationary environments.

5.6. Limitations

We state the boundaries of the present study explicitly. **(i) Attribution faithfulness.** The feature-importance head is not yet causally faithful: under a controlled $+6\sigma$ single-channel spike its top-ranked channel matches the perturbed one only 12.7% of the time (chance 12.5%) and it collapses onto a single channel, so we report it as an architectural placeholder and a future-work target rather than a validated explanation (Section 4.7). **(ii) Effect size.** The IAQ forecasting gain, while statistically significant (Section 4.3), is modest in absolute MSE; we have accordingly reframed the contribution around detection recall and operational guarantees rather than headline accuracy, and we avoid claiming large forecasting improvements. **(iii) Component interactions.** The ablation (Table 4) varies components cumulatively and shows no negative interaction, but it is not a full 2^3 factorial, so synergy versus independence among AUAA, the dynamic adapter, and the attribution head is only partially resolved. **(iv) Domain coverage.** Both evaluation domains (indoor air quality and industrial water circulation) are comparatively stationary; generalization to highly non-stationary or heterogeneous cyber-physical streams, and robustness to adversarial perturbations and inter-sensor synchronization errors (beyond the channel-masking and Gaussian-noise tests of Section 4.8), remain open. **(v) Theoretical scope.** The Monte-Carlo convergence result (Theorem 2) is a standard SLLN/CLT consequence repurposed here to justify the sample budget M rather than a new probabilistic result; its role is to make the $O(M^{-1/2})$ trade-off explicit, and the Chebyshev bound, though valid and empirically respected (Table 12), is intentionally conservative for near-Gaussian scores.

6. Conclusions

We introduced an uncertainty-aware TCN framework that treats predictive uncertainty as an internal architectural signal rather than a post-hoc add-on, accompanied by two distribution-free theoretical guarantees that anchor the operational pipeline. The framework is motivated by three sensor-specific challenges, heterogeneous reliability drift across sensor types, context-dependent anomaly scoring requirements, and the need for actionable per-sensor diagnostic output—and integrates two uncertainty-driven components, trained under a composite objective (a reconstruction loss together with a dynamic-weight regularizer and an attribution loss): (1) the Adaptive Uncertainty-Aware Attention (AUAA) mechanism gates temporal attention by per-sensor predictive uncertainty from MC dropout; and (2) the Dynamic Weight Adapter learns context-sensitive blending of reconstruction error and uncertainty. The Chebyshev-type bound (Theorem 1) furnishes a distribution-free false-positive guarantee on the hybrid score, and the Monte-Carlo convergence result (Theorem 2) sets the sample-size budget M for the inference pass. The architecture also includes a per-sensor attribution head, but we audit it rather than claim it as a result.

Comprehensive experiments on four-month IAQ sensor data demonstrated a 25.2% MSE reduction on the held-out test split (Base 2.20×10^{-4} vs. Full Enhanced 1.65×10^{-4}; R^2 0.9988, SMAPE 11.07%). The ablation isolates the source of this gain: the auxiliary objectives (dynamic-weight regularizer and feature-attribution loss) deliver the dominant reduction, while AUAA contributes uncertainty-gated attention at scoring time without further reducing forecasting MSE. A cross-dataset validation on the Skoltech Anomaly Benchmark (Section 4.5) supplies independent, third-party ground-truth labels and confirms that the Full Enhanced configuration recovers an 8.8× higher F1 (0.477 vs. 0.054) and 14.4× higher recall (0.418 vs. 0.029) than the Base TCN on a domain (industrial water-circulation sensors) disjoint from the IAQ training distribution. The architecture additionally produces a per-sensor attribution output, although a controlled-perturbation audit (Section 4.7) shows this head is not yet causally faithful—we report it transparently as a limitation and future-work target rather than a validated explanation. The Chebyshev bound of Theorem 1 guarantees

$P(\mathrm{FP}) \leq 1/k^2$ without distributional assumptions and is empirically respected on the benchmark. With sub-millisecond per-window inference (0.16 ms at $M = 20$, including the uncertainty pass) and $\sim 15\%$ parameter overhead, the framework is suitable for real-time deployment in smart buildings, environmental monitoring, and industrial process control.

Operational Guarantee: Applying the Chebyshev inequality to anomaly score distributions yields a distribution-free false-positive-rate bound: for any score distribution with finite variance and threshold $\tau = \bar{s} + k \cdot \sigma_s$, the detector satisfies $P(\mathrm{FP}) \leq 1/k^2$. This enables principled threshold selection without parametric distributional assumptions, providing an operational guarantee that complements the empirical percentile-based approach.

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Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Kumar, P.; Khare, M.; Harrison, R.M. A Comprehensive Review of Indoor Air Quality Monitoring Systems and Their Applications. *Science of the Total Environment* **2022**, *828*, 154419. https://doi.org/10.1016/j.scitotenv.2022.154419.
2. Szczurek, A.; Maciejewska, M.; Pietrucha, T. Indoor Air Quality and Prediction of Air Pollution in Rooms Using Artificial Neural Networks. *Building and Environment* **2020**, *177*, 106906. https://doi.org/10.1016/j.buildenv.2020.106906.
3. Coleman, J.R.; Meggers, F. Sensing of Indoor Air Quality—Characterization of Spatial and Temporal Pollutant Evolution Through Distributed Sensing. *Front. Built Environ.*, *4*, 28. https://doi.org/10.3389/fbuil.2018.00028.
4. Abid, K.; Li, J.; Hussain, I.; Ali, H.; Niaz, M. Deep Learning Approach for Indoor Air Quality Prediction Using Multi-Sensor Data. *Building and Environment* **2021**, *197*, 107836. https://doi.org/10.1016/j.buildenv.2021.107836.
5. Chandola, V.; Banerjee, A.; Kumar, V. Anomaly Detection: A Survey. *ACM Computing Surveys* **2009**, *41*, 1–58.
6. Aggarwal, C.C. *Outlier Analysis*, 2nd ed.; Springer, 2017. https://doi.org/10.1007/978-3-319-47578-3.
7. Pang, G.; Shen, C.; Cao, L.; Hengel, A.V.D. Deep Learning for Anomaly Detection: A Review. *ACM Computing Surveys* **2021**, *54*, 1–38.
8. Choi, Y.; Lee, S.; Kim, D. Real-time Anomaly Detection in Indoor Air Quality Sensor Networks Using Deep Autoencoders. *Building and Environment* **2024**, *258*, 111578. https://doi.org/10.1016/j.buildenv.2024.111578.

9. Wei, Y.; Jang-Jaccard, J.; Xu, W.; Sabrina, F.; Camtepe, S.; Boulic, M. LSTM-Autoencoder-Based Anomaly Detection for Indoor Air Quality Time-Series Data. *IEEE Sensors Journal* **2023**, *23*, 3917–3928. https://doi.org/10.1109/JSEN.2022.3232532.
10. Sánchez-Ferrera, A.; Calvo, B.; Lozano, J.A. A Review on Self-Supervised Learning for Time Series Anomaly Detection: Recent Advances and Open Challenges, [2501.15196 [stat]]. https://doi.org/10.48550/arXiv.2501.15196.
11. Ghimire, S.; Deo, R.C.; Casillas-Pérez, D.; Salcedo-Sanz, S.; Acharya, R.; Dinh, T. Electricity demand uncertainty modeling with Temporal Convolution Neural Network models. *Renewable and Sustainable Energy Reviews*, *209*, 115097. https://doi.org/10.1016/j.rser.2024.115097.
12. Cui, P.; Deng, Z.; Hu, W.; Zhu, J. SDE-HNN: Accurate and Well-Calibrated Forecasting Using Stochastic Differential Equations. *ACM Trans. Knowl. Discov. Data*, *19*, 1–23. https://doi.org/10.1145/3691346.
13. Xu, J.; Wu, H.; Wang, J.; Long, M. Anomaly Transformer: Time Series Anomaly Detection with Association Discrepancy. In Proceedings of the Proceedings of the 10th International Conference on Learning Representations (ICLR), 2022.
14. Bai, S.; Kolter, J.Z.; Koltun, V. An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling. *arXiv preprint arXiv:1803.01271* **2018**.
15. Gal, Y.; Ghahramani, Z. Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning. *JMLR: W&CP* **2016**, *48*.
16. Hochreiter, S.; Schmidhuber, J. Long Short-Term Memory. *Neural Computation* **1997**, *9*, 1735–1780. https://doi.org/10.1162/neco.1997.9.8.1735.
17. Cho, K.; van Merriënboer, B.; Gulcehre, C.; Bahdanau, D.; Bougares, F.; Schwenk, H.; Bengio, Y. Learning Phrase Representations Using RNN Encoder–Decoder for Statistical Machine Translation. In Proceedings of the Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2014, pp. 1724–1734.
18. van den Oord, A.; Dieleman, S.; Zen, H.; Simonyan, K.; Vinyals, O.; Graves, A.; Kalchbrenner, N.; Senior, A.; Kavukcuoglu, K. WaveNet: A Generative Model for Raw Audio. *arXiv preprint arXiv:1609.03499* **2016**.
19. Zhang, Q.; Wu, J.; Yang, C.; Pan, T.; Yu, X. A Review of the Application of Temporal Convolutional Networks for Time Series Forecasting. *Engineering Applications of Artificial Intelligence* **2023**, *102*, 104252. https://doi.org/10.1016/j.engappai.2023.104252.
20. Thill, M.; Konen, W.; Wang, H.; Bäck, T. Temporal convolutional autoencoder for unsupervised anomaly detection in time series. *Applied Soft Computing*, *112*, 107751. https://doi.org/10.1016/j.asoc.2021.107751.
21. Wang, M.; Qin, F. A TCN-Linear Hybrid Model for Chaotic Time Series Forecasting. *Entropy*, *26*, 467. https://doi.org/10.3390/e26060467.
22. Mehta, A.; Yang, W. NAC-TCN: Temporal Convolutional Networks with Causal Dilated Neighborhood Attention for Emotion Understanding. In Proceedings of the Proceedings of the 2023 7th International Conference on Video and Image Processing, 2023, pp. 9–16, [2312.07507 [cs]]. https://doi.org/10.1145/3639390.3639392.
23. Audibert, J.; Michiardi, P.; Guyard, F.; Marti, S.; Zuluaga, M.A. USAD: UnSupervised Anomaly Detection on Multivariate Time Series. In Proceedings of the Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD), 2020, pp. 3395–3404.
24. Malhotra, P.; Ramakrishnan, A.; Anand, G.; Vig, L.; Agarwal, P.; Shroff, G. LSTM-based Encoder-Decoder for Multi-sensor Anomaly Detection. In Proceedings of the ICML 2016 Anomaly Detection Workshop, 2016.
25. Tuli, S.; Casale, G.; Jennings, N.R. TranAD: Deep Transformer Networks for Anomaly Detection in Multivariate Time Series Data. *Proceedings of the VLDB Endowment* **2022**, *15*, 1201–1214.
26. Zhou, H.; Zhang, S.; Peng, J.; Zhang, S.; Li, J.; Xiong, H.; Zhang, W. Informer: Beyond Efficient Transformer for Long Sequence Time-Series Forecasting. In Proceedings of the Proceedings of the AAAI Conference on Artificial Intelligence, 2021, Vol. 35, pp. 11106–11115.
27. Wu, H.; Hu, T.; Liu, Y.; Zhou, H.; Wang, J.; Long, M. TimesNet: Temporal 2D-Variation Modeling for General Time Series Analysis. In Proceedings of the Proceedings of the International Conference on Learning Representations (ICLR), 2023.

28. Zhao, H.; Wang, Y.; Duan, J.; Huang, C.; Liu, D.; Wang, J.; Sun, T.; Li, Z.; Liu, J.; Yao, H. Multivariate Time-Series Anomaly Detection via Graph Attention Network. In Proceedings of the IEEE International Conference on Data Mining (ICDM), 2020, pp. 841–850.
29. Kendall, A. Uncertainties in Deep Learning. PhD thesis, University of Cambridge, 2017.
30. Srivastava, N.; Hinton, G.; Krizhevsky, A.; Sutskever, I.; Salakhutdinov, R. Dropout: A Simple Way to Prevent Neural Networks from Overfitting. *Journal of Machine Learning Research (JMLR)* **2014**, *15*, 1929–1958.
31. Lakshminarayanan, B.; Pritzel, A.; Blundell, C. Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles. In Proceedings of the Advances in Neural Information Processing Systems (NeurIPS), 2017, Vol. 30, pp. 6402–6413.
32. Vaswani, A.; Shazeer, N.; Parmar, N.; Uszkoreit, J.; Jones, L.; Gomez, A.N.; Kaiser, Ł.; Polosukhin, I. Attention is all you need. In Proceedings of the Advances in Neural Information Processing Systems, 2017, Vol. 30, pp. 5998–6008.
33. Angelopoulos, A.N.; Bates, S. A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification. *arXiv preprint arXiv:2107.07511* **2021**.
34. Boucheron, S.; Lugosi, G.; Massart, P. *Concentration Inequalities: A Nonasymptotic Theory of Independence*; Oxford University Press: Oxford, UK, 2013.
35. Kendall, A.; Gal, Y.; Cipolla, R. Multi-Task Learning Using Uncertainty to Weigh Losses for Scene Geometry and Semantics. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* **2018**, pp. 7482–7491.
36. Wu, R.; Keogh, E.J. Current Time Series Anomaly Detection Benchmarks are Flawed and are Creating the Illusion of Progress. *IEEE Transactions on Knowledge and Data Engineering* **2023**, *35*, 2421–2429. https://doi.org/10.1109/TKDE.2021.3112126.
37. Lai, K.H.; Zha, D.; Xu, J.; Zhao, Y.; Xu, G.; Hu, X. Revisiting Time Series Outlier Detection: Definitions and Benchmarks. In Proceedings of the NeurIPS Datasets and Benchmarks Track, 2021.
38. Hundman, K.; Constantinou, V.; Laporte, C.; Colwell, I.; Soderstrom, T. Detecting Spacecraft Anomalies Using LSTMs and Nonparametric Dynamic Thresholding. In Proceedings of the Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining. ACM, 2018, pp. 387–395. https://doi.org/10.1145/3219819.3219845.
39. Su, Y.; Zhao, Y.; Niu, J.; Liu, R.; Sun, W.; Pei, D. Robust Anomaly Detection for Multivariate Time Series through Stochastic Recurrent Neural Network. In Proceedings of the Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD), 2019, pp. 2828–2837.
40. Goh, J.; Adepu, S.; Junejo, K.N.; Mathur, A. Anomaly detection in cyber-physical systems using recurrent neural networks. In Proceedings of the 2017 IEEE 18th International Symposium on High Assurance Systems Engineering (HASE). IEEE, 2017, pp. 140–145.
41. Mathur, A.P.; Tippenhauer, N.O. SWaT: A Water Treatment Testbed for Research and Training on ICS Security. In Proceedings of the 2016 International Workshop on Cyber-physical Systems for Smart Water Networks (CySWater). IEEE, 2016, pp. 31–36. https://doi.org/10.1109/CySWater.2016.7469060.
42. Filonov, P.; Lavrentyev, A.; Vorontsov, A. Multivariate Industrial Time Series with Cyber-Attack Simulation: Fault Detection Using an LSTM-based Predictive Data Model. In Proceedings of the NIPS Time Series Workshop, 2016. Origin of the Skoltech water-circulation testbed reused for SKAB.
43. Katser, I.D.; Kozitsin, V.O. Skoltech Anomaly Benchmark (SKAB). https://github.com/waico/SKAB, 2020. Version 0.9; multivariate sensor time-series with point-anomaly and change-point labels from an industrial water-circulation testbed at Skoltech, https://doi.org/10.34740/kaggle/dsv/1693952.
44. Ester, M.; Kriegel, H.P.; Sander, J.; Xu, X. A Density-Based Algorithm for Discovering Clusters in Large Spatial Databases with Noise. In Proceedings of the Proceedings of the 2nd International Conference on Knowledge Discovery and Data Mining (KDD-96). AAAI Press, 1996, pp. 226–231.

45. He, K.; Zhang, X.; Ren, S.; Sun, J. Deep Residual Learning for Image Recognition. In Proceedings of the Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016, pp. 770–778.
46. Kingma, D.P.; Ba, J. Adam: A Method for Stochastic Optimization. In Proceedings of the Proceedings of the 3rd International Conference on Learning Representations (ICLR), 2015.
47. Ioffe, S.; Szegedy, C. Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift. *Proceedings of the 32nd International Conference on Machine Learning (ICML)* **2015**, *37*, 448–456.
48. Liu, F.T.; Ting, K.M.; Zhou, Z.H. Isolation Forest. *Proceedings of the IEEE International Conference on Data Mining (ICDM)* **2008**, pp. 413–422. https://doi.org/10.1109/ICDM.2008.17.
49. Park, D.; Hoshi, Y.; Kemp, C.C. A Multimodal Anomaly Detector for Robot-Assisted Feeding Using an LSTM-Based Variational Autoencoder. *IEEE Robotics and Automation Letters* **2018**, *3*, 1544–1551.
50. Wang, Y.; Zhang, L.; Liu, H. Res2coder: A Two-Stage Residual Autoencoder for Unsupervised Multivariate Time-Series Anomaly Detection. *Applied Intelligence* **2025**. Compact reimplementation used as a unified-protocol baseline in this work.